



OEBB DOSTO NEU PROJECT NOMAD CONNECT RELEASE 2025.2.1

Test Exit Report

16 April 2026

PURPOSE

To provide evidence of specified SW build based on predefined test criteria.

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Business Management System Standards



The Business Management System of Nomad Digital Ltd has been certified to ISO 9001:2015.

This standard is applicable to management activities associated with the design, development, delivery and service support of connectivity and data solutions for the rail and transport sector.

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Preface

Document Guidelines

The following notices and statements are used in this document:



Draws your attention to particularly important information on handling the product, the product itself or to a particular part of the documentation.

Indicates that if the proper precautions are not taken, this can result in product damage.



Emphasizes or supplements important points of the main text. Helps users apply the techniques and procedures described in the text suggesting alternative methods that may not be obvious.

Reporting Issues with This Document

While reasonable precautions have been taken, technical inaccuracies or typographical errors may appear in the document. Document users can report issues through their project representative.

1. Introduction

1.1. Glossary

Term	Description
IOB	Internet on Board
CCU	Communications Control Unit
AP	Access Point
MEN	Mikro Elektronik GmbH
WCB	Wireless Client Bridge
WAN	Wireless Access Network
GPS	Global Positioning System
QA	Quality Assurance
NC	Nomad Connect
SW	Switch
BQ	Big Query Collector and related services
Global IP	The IP within the Nomad Network that the CCU can be accessed via
CDC	Customer Data Centre
RTPi	Real Time Passenger Information

1.2. Introduction

The purpose of this Test Exit Report is to provide a concise overview of the functional testing of the *2025.2.1 Nomad Connect Build* on the OEGB Dosto Neu test Bench.

The completed document will provide key Stakeholders with an understanding of 'what' functional tests were performed; and of the 'why' and 'how' the functional testing in relation to the fleet specific build prior to the fleetwide release of this software.

September 2025:

The purpose of this tasting phase is to provide a targeted validation of core functionality after upgrading the NC version to 2025.2.1 - for details please see point 2.2 - Items Tested; September 2025 section.

2. Scope

2.1. Scope

These tests are intended to provide final assurance of the proposed release of Nomad Connect for the DEL-OBB-035 IoB Bench.

The Nomad Digital IoB solution for OEGB Dosto Neu project consists of multiple devices installed on board the vehicle to facilitate passenger, staff or third-party network access. The centralised train to shore link uses the Nomad R5001C communications control unit (CCU) to construct cellular or wireless connections to the wayside infrastructure and establish a secure tunnel connection to the customers data centre (CDC).

The IoB test exit report lists the relevant functionality tests to be performed on the Nomad system internal test bench, their expected outcomes and results evidence. The test bench is used to validate as many of the systems as possible, however this is dependent on the CDC infrastructure available at the time and any in country specifics.

The software package tested with the scope of the test exit report will become the basis for the first of class vehicle. Any subsequent changes to the build after that point would be controlled within the project release note documentation pack.

The tests in this report were carried out on the bench systems identified in section 4.2.1

The testing is intended as functional testing post QA, to identify fleet specific test cases and functional requirements outside of the scope of QA testing.

2.2. Items Tested

The following items were tested within the Test exit report, this list is not exhaustive but should be considered an overview of the items tested.

2.2.1. Basic system functionality

2.2.2. System Boot Functionality

- System services

- 2.2.3. Tunnel Connectivity
- 2.2.4. OBN Functionality
- 2.2.5. OBN Updating functionality
- 2.2.6. OBN Validation functionality
- 2.2.7. WAN functionality
- 2.2.8. On-board networking functionality
- 2.2.9. CCU APIs
- 2.2.10. System Update functionality
- 2.2.11. GPS functionality
- 2.2.12. Passenger Functionality test
- 2.2.13. Portal Functionality
- 2.2.14. In relation to general functionality
- 2.2.15. On-board monitoring services
- 2.2.16. Including on-train Zabbix
- 2.2.17. MQTT Reporting
- 2.2.18. Zabbix Functionality
- 2.2.19. NMS Reporting

17 September 2025 - NC upgrade to 2025.2.1 core tests

- 6.3.1.3 - System boot and Firmware Validation test
- 6.3.1.4 - Show running processes and CPU utilization
- 6.3.1.5 - Boot Resilience and Initialization
- 6.3.1.7 - CCU can form active tunnel connection
- 6.3.1.8 - NTP synchronisation
- 6.3.2.1 - OBN Automatic discovery
- 6.3.2.3 - OBN Successful update of device configuration
- 6.3.4.1 - Switch Functionality
- 6.3.4.2 - AP Functionality
- 6.3.4.3 - DHCP and User Session Counting
- 6.3.4.4 - AP Accessibility
- 6.3.5.1 - Ensure GPS API calls to localhost are possible
- 6.3.5.2 - Ensure Train Identification API calls to localhost are possible
- 6.3.5.3 - Ensure Train captive portal API calls to localhost are possible
- 6.3.5.4 - Ensure Engineering Pages are available
- 6.3.8.3 - Passenger Connectivity Flow Using Windows Laptop/Desktop
- 6.3.8.6 - Passenger DHCP Functionality
- 6.3.8.8 - DNS Functionality
- 6.3.8.9 - DNS response time

2.3. Out of Scope

The following specifies areas that will not be covered during testing. These items are non-project specific items that fall within the purview of QA, fall under separate test reports, or are not a feature altered by Devops.

- NMS
 - With relation to how the train is represented and other general functionality and not data reporting
- Zabbix
 - With relation to server-side Zabbix configuration and performance
 - With relation to the train configuration
- Fleetview
 - With relation to reports and their appearance
 - How data is reported
- RTPI
- Information Security testing
 - To be carried out in a separate test
- Portal Performance
 - In relation to functionality that is within the purview of ND Hildesheim team

2.4. Limitations of testing

The following highlights explain any limitations that occurred in the execution of the TER; what was done to overcome them, what had to be mitigated, and what could not be mitigated or overcome.

Insufficient hardware allocation – The project provided enough equipment to build a 4 car consist, because of this Dev-Ops were limited in how much they could test. This means that Dev-Ops cannot guarantee that the six-car-configuration will work with no noticeable issues. However, in this case, some preliminary testing has already been carried out by the R&D team to ensure 6 car configuration compatibility.

SIM provisioning – This is generally a limitation of the TOC or the in-country SIM provider and occurs in most Dev-Ops TERs. SIM tests should be performed during in-country testing to mitigate this limitation. DevOps further mitigated this limitation by using non-standard SIMs to ensure that modem operation of the build behaved as expected.

3. Testing Methodology

Testing phases that will be used for this release are as follows:

- DevOps bench testing in Newcastle:
 - 1.1. Pipe Test: Exploratory validation test to assess if the release and environment in general is in a stable/suitable state for formal test activities to commence.
 - Functional Integration and Validation Testing.

4. Environment Requirements

4.1. Hardware Platforms

The NC 2025.2.1 Release was tested against a representative DEL-OB-035 hardware loadout:

The same setup is used to test the 2025.2.1 Release update.

Newcastle Bench:

CCU & other related hardware	Switch	Access Point	Modem
R5001C CCU	VDS Rail (x12)	Westermo (x3)	LM990 Telit modem (x4)

The Newcastle bench hardware had the following hardware:

Hardware	Firmware	Configuration Version	Hardware Revision	Notes
R5001C CCU	2025.2.1	7ce1d29ce6	C	
BIOS	1.02.10			
VDS Rail	7.4.2			
Westermo	6.11.2-0			
LM990 Telit	M0R.000002		A28	

4.2. Bench Definitions

As part of testing procedure, benches are used to test releases and project specific configurations. Whilst the hardware used on the test bench is the same as that used on the trains, there are some differences regarding physical limitations of the test area, cabling, etc. The following covers benches used in testing and how they differ from the final production units.

Note: Benches do not try to replicate or take into consideration Wi-Fi performance found on train (i.e., How the shape of carriages affects signal), power supplies or how any environmental factors such as speed, temperature, and airflow affect its operation.

4.2.1. Newcastle Bench

The Newcastle DEL-OB-035 bench (known as T112) is a Nomad facing Bench located with the Newcastle Head office of Nomad Digital. The setup aims to replicate a two-car setup of the final solution with both CCUs and ring network. The following differs from the final production systems:

- SIMs:
 - SIM cards may not be used in all tests (in favour of ethernet backbone)
 - Where SIMs are used, they are UK SIMs
- USB Ethernet interface
 - Used only in throughput tests
 - To act in place of SIM cards (WANs)
 - Ethernet has 1Gbps capabilities
- Other factors
 - The CCU routes through the internal office network of Newcastle and will be dealing with traffic congestion of the office
 - GPS is not plugged in consistently

- Access Points
 - SSIDs are split by radio for improved diagnostics
 - Hidden SSIDs made visible
 - Staff SSID renamed to include fleet name for clarity

Bench Equipment

The following lists all equipment on the Newcastle Bench, along with internal ID which can be used for further information.

Equipment	Devops asset code
CCU 1	/
SW 1	/
SW 2	/
SW 3	/
SW 4	/
SW 5	/
SW 6	/
SW 7	/
SW 8	/
SW 9	/
SW 10	/
SW 12	/
AP 1	/
AP 2	/
AP 3	/
Switch PSU 1	/
Switch PSU 2	/
Switch PSU 3	/
CCU PSU	/

4.3. Software Platforms

Software specifications for the CCU and Portal are listed below:

Note that the configuration ID refers to a unique hash of the configuration as it stands at the time of testing.

CCU Build	Configuration ID	Portal
Nomad Connect 2024.3.1	8d598dd894e4d7c40bf32d473cbafa50ce0bd286	nde:2023Q4-10.1.7-x64
Nomad Connect 2025.2.1	7ce1d29ce65f451a92d5bdd51155d4fdaded304c	nde:2023Q4-10.1.7-x64

Software package breakdown of the Nomad Connect Build which were used specifically for the project:

Package	Version
21net-security-dostoneu	0.0.6
nd-obn-template-dostoneu-nv4	0.0.2

For the full release notes of Nomad Connect 2025.2.1 please see: <https://nomad-digital.atlassian.net/wiki/spaces/2SD/pages/4710760468/Nomad+Connect+2025.2+Release+Notes>

4.4. Client Test Devices

The following devices were used for client facing functionality testing.

Device	OS	Browser
Windows Virtual Machine	Windows 10- 19042.1288	Edge (chromium based)
Linux Virtual Client	Ubuntu	Firefox (Gecko)
Android Mobile	Android 11.0.0.190	Chrome (chromium based)
iPhone mobile	IOS 15.4.1	Safari (WebKit)

Several devices were used for testing to ensure a range of compatibility for end users; further, some devices were used due to diagnostic and debugging tools to give greater aCCUracy in testing.

The Following devices were also used for specifically testing external facing items (I.e., looking at NMS or checking Engineer pages via global IP)

Device	OS	Browser
Linux Desktop	Ubuntu 24.04.02 LTS - 5.4.0-107-generic	Brave (Chromium based)

5. Resources and Responsibilities

Activity	Responsibility
Preparation of test plan, test cases and requirement coverage	DevOps
Set up test data and environment	DevOps
Test execution & defect reporting	DevOps
Test Exit Report	DevOps

6. Test Scenarios & Outcome

6.1. Test Overview

- Analysis of the functional requirements was conducted, and the appropriate Test Cases developed.
- The intention of the test cases is to prove the required functionality operates as intended and validates adherence to the project design parameters.

Completed tests follow a highlighted scheme:

Colour	Meaning
	The test was completed successfully
	The test could not be completed in this phase of testing
	The test failed due to an issue unrelated to the build and would otherwise be considered successful
	The test failed due to fundamental flaws in the build or configuration
	Basic functionality, verified in previous testing. Not tested in this TER.

6.2. Test Scenarios

The test scenarios created using the bench equipment are intended to replicate, as much as possible, the real-life situations users of the system will experience. They will also be testing, where applicable, remote connectivity, health monitoring and reporting features.

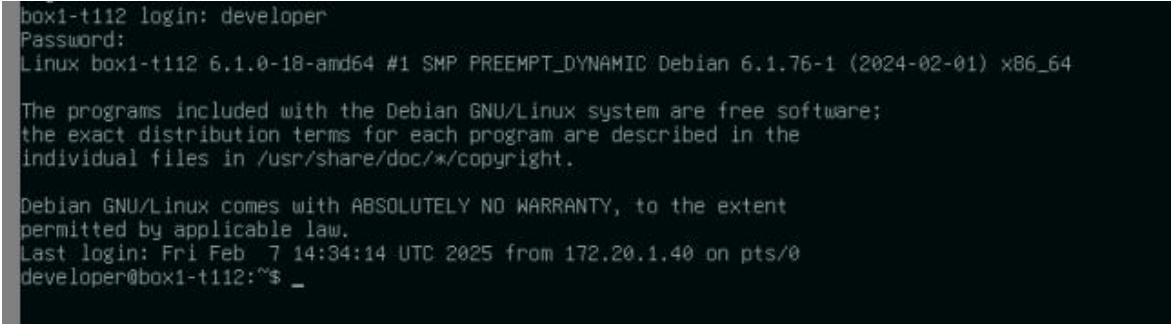
6.3. Test Results & Outcome

Test cases along with the execution steps taken and results evidence to justify the final pass state.

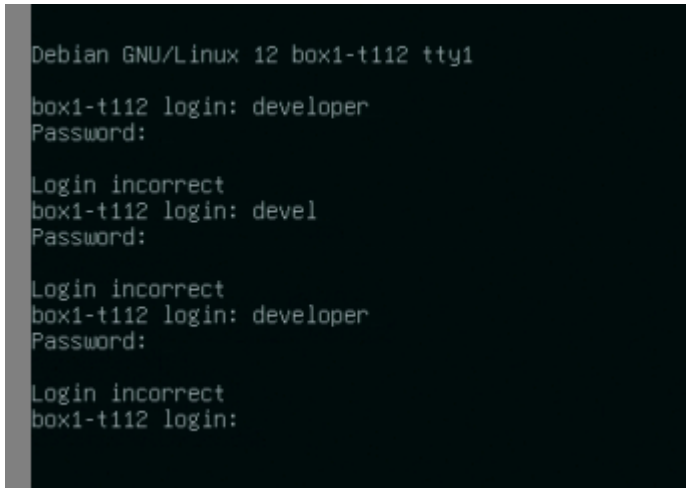
6.3.1. Basic Functionality tests

6.3.1.1. Serial connectivity

Test ID:	6.3.1.1
Description:	Test the Serial connectivity functionality. Ensure that Baud rate is set to 115200, access is protected by username and password
Execution Steps	Ensure CCU is powered, that Status light on the CPU Blade/Chassis flashes Connect a usb male-to-male (or other applicable serial cable) between CCU and your device Check 'dmesg'/device manager for the connection Linux: run <i>cmd 'dmesg grep tty'</i> Windows: <i>Check Device Manager for Com port</i> Confirm that the device is recognised as a serial cable Linux: <i>dmesg states serial adapter</i> Windows: <i>Device manager shows device as com port</i> Open a connection to the serial interface using tool of choice Enter the username and password Access to the CCU should now be granted
Expected Outcome:	Once the USB male-to-male console cable (or other applicable serial cable) is connected between the CCU and a common interface device, it will be detected by the common interface device as a serial com (FTDI) device. Accessing it will require serial rate 115200. To access CCU functionality will require a username and password.
Preconditions:	CCU BIOS is unmodified Functional, pre-staged image installed
Bench:	Devops

Test Method:	Functional Testing
Test Device:	Windows
Execution Date:	07/02/2025
Executer:	Szymon Dutkiewicz
Evidence:	
 <pre> box1-t112 login: developer Password: Linux box1-t112 6.1.0-18-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.76-1 (2024-02-01) x86_64 The programs included with the Debian GNU/Linux system are free software; the exact distribution terms for each program are described in the individual files in /usr/share/doc/*/copyright. Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent permitted by applicable law. Last login: Fri Feb 7 14:34:14 UTC 2025 from 172.20.1.40 on pts/0 developer@box1-t112:~\$ _ </pre>	

Test ID:	6.3.1.1b
Description:	Test the Serial connectivity is only possible with the predefined user/pass pair
Execution Steps	1. Ensure CCU is powered, that Status light on the CPU Blade/Chassis flashes 2. Connect a usb male-to-male (or other applicable serial cable) between CCU and your device Check 'dmesg'/device manager for the connection Linux: run <i>cmd 'dmesg grep tty'</i> Windows: <i>Check Device Manager for Com port</i> Confirm that the device is recognised as a serial cable Linux: <i>dmesg states serial adapter</i> Windows: <i>Device manager shows device as com port</i> 3. Open a connection to the serial interface using tool of choice 4. Try logging into the CCU with incorrect details a) user: developer / password: test

	b) user: devel / password: *actual console password* c) user: developer / password: "blank" d) user: root / password: root e) user: root1 / password: *actual console password* f) user: root / password: "blank"
Expected Outcome:	Access to the CCU via serial is not possible with incorrect username and/or password
Preconditions:	CCU BIOS is unmodified Functional, pre-staged image installed
Bench:	Devops
Test Method:	Negative Testing
Test Device:	Windows
Execution Date:	07/02/2025
Executer:	Szymon Dutkiewicz
Evidence:  <pre> Debian GNU/Linux 12 box1-t112 tty1 box1-t112 login: developer Password: Login incorrect box1-t112 login: devel Password: Login incorrect box1-t112 login: developer Password: Login incorrect box1-t112 login: </pre>	

6.3.1.2. SSH connectivity

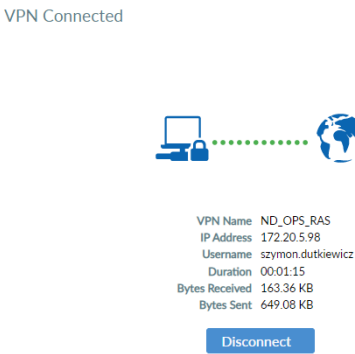
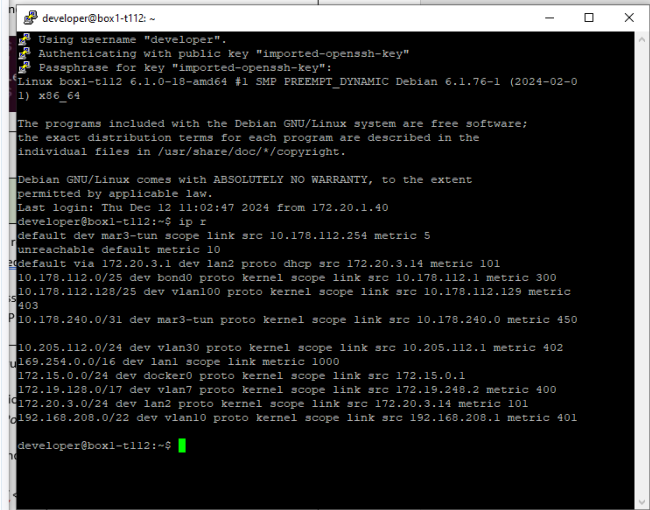
Test ID:	6.3.1.2
Description:	Ensure that the CCU can be remotely accessed via the ssh protocol The CCU should be accessible via port 22 and with user developer either via the local address or global IP
Execution Steps	Open a terminal or application capable of running openssh Applications can be putty, <i>PowerShell</i> or <i>Linux subsystem</i> Enter the following command substituting the <IP> for the IP of the bench and <key> for your private key <code>ssh developer@<IP> -p 22 -i <key></code> <i>NOTE: This will vary given the application</i> Once the command is run a ssh connection should be formed and a terminal prompt on the CCU should be available
Expected Outcome:	The CCU is accessible with ssh via port 22 and its local address or global address
Preconditions:	Previous tests passed, connectivity to CCU in form of local network or MAR tunnel
Bench:	Devops Bench
Test Method:	Functional Testing
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	

```
david@david:~$ ssh developer@10.179.122.1  
sudo sLinux box1-t122 6.1.0-31-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.1.128-1 (2025-02-07) x86_64  
  
Welcome to Linux! Type man to learn more, or press Ctrl+D to continue.  
  
Last login: Wed Nov 12 08:28:40 2025 from 172.20.5.87  
developer@box1-t122:~$ sudo su
```

Test ID:	6.3.1.2b
Description:	Ensure that the CCU cannot be remotely accessed via the ssh protocol when incorrect details used. The CCU should not be accessible via port 22 and with user developer either via the local address or global IP when the RSA key is not on the unit
Execution Steps	Open a terminal or application capable of running openssh Applications can be putty, PowerShell or Linux subsystem Enter the following command substituting the <IP> for the IP of the bench and <key> for your private key ssh unknownUser@<IP> -p 22 -i <key> NOTE: This will vary given the application Once the command is run an ssh connection begin to form but be rejected due to either a mismatch of key or username
Expected Outcome:	The CCU is accessible with ssh via port 22 and its local address or global address
Preconditions:	Previous tests passed, connectivity to CCU in form of local network or MAR tunnel
Bench:	Devops Bench
Test Method:	Negative Testing
Test Device:	Linux
Execution Date:	12/11/2025

Executer:	Davud Zejnelovic
Evidence:	
<pre>davud@davud:~\$ ssh admin@10.179.122.1 admin@10.179.122.1: Permission denied (publickey). davud@davud:~\$ ssh root@10.179.122.1 root@10.179.122.1: Permission denied (publickey). davud@davud:~\$</pre>	

Test ID:	6.3.1.2c
Description:	Ensure that the CCU can be remotely accessed via ssh using the nomad VPN when correct details are used The CCU should not be accessible via port 22 and with user developer either via the local address or global IP when the RSA key is not on the unit
Execution Steps	Ensure that Nomad VPN is running (openvpn does not count) Open a terminal or application capable of running openssh Applications can be putty, PowerShell or Linux subsystem Enter the following command substituting the <IP> for the IP of the bench and <key> for your private key ssh developer@<IP> -p 22 -i <key> <i>NOTE: This will vary given the application</i> Once the command is run an ssh connection should be formed and a terminal prompt on the CCU should be available
Expected Outcome:	The CCU is accessible with ssh via port 22 and its local address or global address
Preconditions:	Previous tests passed, connectivity to CCU in form of local network or MAR tunnel

	A VPN Tunnel
Bench:	Devops Bench
Test Method:	Negative Testing
Test Device:	Windows
Execution Date:	12/12/2024
Executer:	Szymon Dutkiewicz
Evidence: VPN Connected  	

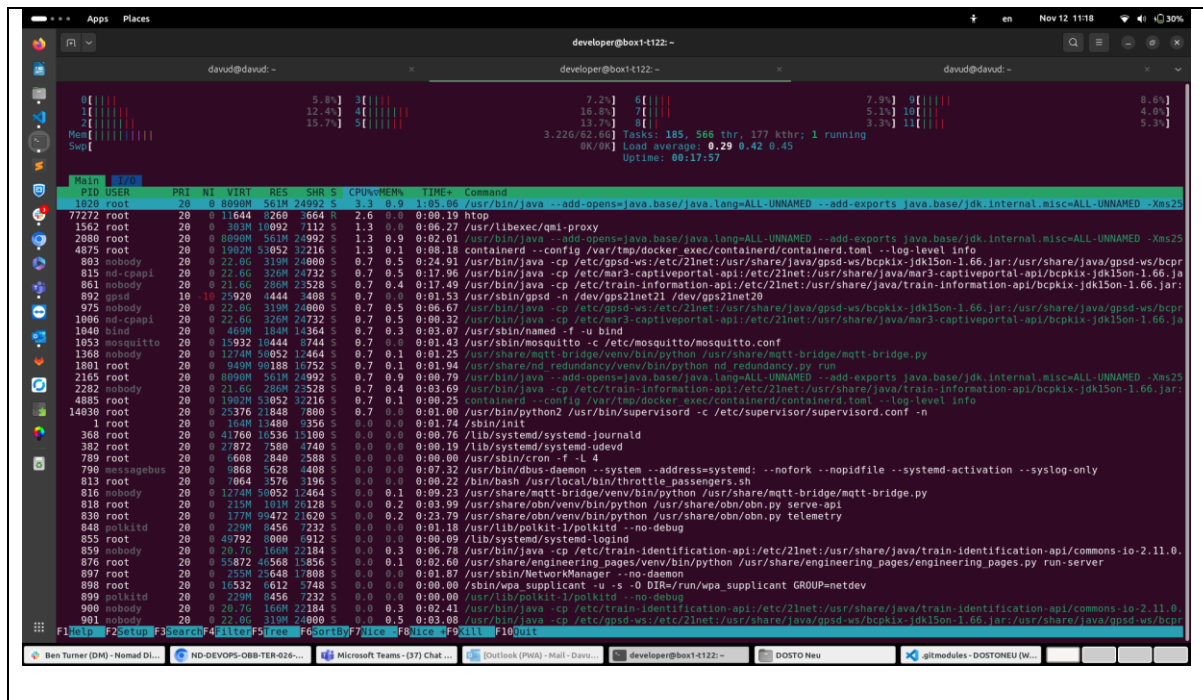
6.3.1.3. System Boot and Firmware validation test

Test ID:	6.3.1.3
Description:	Ensure system boots with no errors and firmware version can be validated. For this test we simply power on the system while connected via the console port.
Execution Steps:	Connect a serial and interface device to the CCU Open a serial connection to the CCU Detailed steps can be found in 6.3.1.1 Power cycle the CCU

	Observe boot sequence and ensure no critical issues on boot Using serial login to the CCU Run command <i>'cat /.snapshot/state'</i>
Expected Outcome:	System powered on and boots without issue (no visible issues in boot sequence) The version of the system can be viewed via <i>.snapshots/state</i> Snapshot state shows version as hash beginning with hash mentioned in section 3.3 (6d9d94e093)
Preconditions:	System image installed Console access to the CCU Serial capabilities (6.3.1.1) confirmed working
Bench:	Devops Bench
Test Method:	Functional testing
Test Device:	Linux
Execution Date:	17/09/2025
Executer:	Piotr Romanowski
Evidence: <pre> root@box1-t121:/home/developer# cat /.snapshots/state KNOWN_WORKING_VERSION=8d598dd894e4d7c40bf32d473cbafa50ce0bd286 VERSION=8d598dd894e4d7c40bf32d473cbafa50ce0bd286 KNOWN_WORKING_TAG=_ TAG=_ KNOWN_WORKING_ENVIRONMENT=dostoneu_migration_mar5 ENVIRONMENT=dostoneu_migration_mar5 KNOWN_WORKING_NOMAD_CONNECT_VERSION=a33b27902c4aeb1d983d99eaac3b93447f1ea1b7 NOMAD_CONNECT_VERSION=a33b27902c4aeb1d983d99eaac3b93447f1ea1b7 KNOWN_WORKING_NOMAD_CONNECT_TAG=2025.2.1 NOMAD_CONNECT_TAG=2025.2.1 FAIL=0 root@box1-t121:/home/developer# </pre>	

6.3.1.4. Show running processes and CPU utilization

Test ID:	6.3.1.4
Description:	Ensure system runs with only necessary system processes, and CPU utilization is acceptable. For this test, we observe the output of the top command for five minutes.
Execution Steps	Access the CCU either via serial or ssh On the CCU console/terminal run command ' <i>top</i> ' Load average (top line) show report three values under two
Expected Outcome:	Top shows no unexpected processes; system CPU resources are largely idle with overall CPU utilization less than 10%.
Preconditions:	CCU has been online for at least five minutes CCU Is not performing any other test
Bench:	Devops Bench
Test Method:	Non-functional testing
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	



6.3.1.5. Boot Resilience and Initialization

Test ID:	6.3.1.5
Description:	Validate over four boots that the core system boots in a consistent and timely manner, no unexpected errors occur relating to the core system and that all modems, switches, and APs are detected consistently upon system initialization
Execution steps	Power cycle the CCU four times leaving a ten-minute gap between power cycles Access the CCU either via ssh or console After the fourth power cycle, confirm that the CCU is connected to the backend using 'marcli' Confirm that marcli shows active connection Cat /.snapshot/state check that configuration hash is the same mentioned in section 3.3
Expected Outcome:	System boots. Marcli shows the modems power up and become active. OBN shows all switches and AP discovered within the OBN discovery cycle

	Try creating a new file: <pre>touch test.txt /home/developer</pre> <pre>touch test.txt /etc/</pre>
Expected Outcome:	You should be unable to make any modifications to the filesystem
Preconditions:	CCU is configured
Bench:	Devops Bench
Test Method:	Robustness Testing
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:  <pre> root@box1-t122:/home/developer# touch test.txt /home/developer touch: cannot touch 'test.txt': Read-only file system touch: setting times of '/home/developer': Read-only file system root@box1-t122:/home/developer# touch test.txt /etc/ touch: cannot touch 'test.txt': Read-only file system touch: setting times of '/etc/': Read-only file system root@box1-t122:/home/developer# date Wed Nov 12 10:21:30 UTC 2025 root@box1-t122:/home/developer# </pre>	

6.3.1.7. CCU can form active tunnel connection

Test ID:	6.3.1.7
Description:	The CCU forms an active tunnel connection through the cellular modems or through the default network interface (local network bridge)
Execution Steps:	Console into CCU Follow steps in first test

	Reboot the CCU Once the CCU comes up use the known username and password Run command '<i>watch marcli</i>' Wait for the modems or default device (eth2) to go into state 'ACTIVE' Once they are in the state 'ACTIVE' Run 'ping -c5 8.8.8.8' The command should come back with 0% packet loss If any loss, repeat step Run command 'curl ifconfig.io' Command should return '185.201.63.20' Note: if on deploy backend
Expected Outcome:	When the CCU boots, within the space of one minute. It can form a connection using the mar tunnel to the backend
Preconditions:	CCU is staged CCU is powered CCU has ethernet, wan link or SIMs
Bench:	Devops
Test Method:	Functional testing
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	

```
root@box1-t122:/home/developer# marcli modem
state      powered  geo-disabled  sim  operator  cell  lac  network  band  iccid
-----
ce0p0: SEARCHING      TRUE      FALSE      1      0      0      0
ce1p0: CONNECTED      TRUE      FALSE      4      23203  10028555  14061  LTE      B3      80430103122049054020
ce2p0: CONNECTED      TRUE      FALSE      1      23201  2063629  17006  LTE      B3      80430103122049054028
ce3p0: CONNECTED      TRUE      FALSE      4      23203  10028555  14061  LTE      B3      80430103122049054020
root@box1-t122:/home/developer# ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=119 time=68.6 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=119 time=31.0 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=119 time=41.9 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=119 time=74.0 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=119 time=46.1 ms
^C
--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 400ms
rtt min/avg/max/mdev = 31.024/52.324/74.016/16.360 ms
root@box1-t122:/home/developer# date
Wed Nov 12 10:22:24 UTC 2025
root@box1-t122:/home/developer#
```

Test ID:	6.3.1.7b
Description:	The CCU forms an active tunnel to specified HA – test all deployed HAs – this ensures all HAs in the backend are configured
Execution Steps:	Configure CCU >vim /etc/mar3/mar3.properties Change the following line to the correct IP: >mar3.client.tunnel.remote.host=77.237.62.210 >service mar3 restart Run command “marcli all” Confirm tunnel IP and run ping Run ping –c 5 8.8.8.8
Expected Outcome:	0-5m delay for OSPF to route through another HAs
Preconditions:	CCU is staged CCU is powered CCU has ethernet, wan link or SIMs
Bench:	Devops
Test Method:	Functional testing
Test Device:	Linux

Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	
Tested: 77.237.62.210, 77.237.62.212, 77.237.62.214, 77.237.62.216, 77.237.62.218	

6.3.1.8. NTP synchronisation

Test ID:	6.3.1.8
Description:	NTP is an important feature of NC, it is of great priority that it syncs with a server
Execution Steps	2. Ensure connection to CCU via SSH or console 3. Run command '<i>date</i>' on CCU to get date 1. Ensure the date is in the correct timezone 2. Confirm that date & time matches current date and time 3. Run command '<i>ntpq -pn</i>' to check synchronisation with public servers 4. Confirm that the column 'offset' is between -2 and +2
Expected Outcome:	This test is passed, when the CCU is synchronized with a NTP-server (date and time report accurately within 2 seconds)
Preconditions:	Tunnel can form Active connection to the CCU (serial or ssh)
Bench:	Devops Bench
Test Method:	Functional Testing
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic

Evidence:

```

root@box1-t122:/home/developer#
root@box1-t122:/home/developer# ntpq -pn
remote                                refid                                st t when poll reach  delay  offset  jitter
=====
0.debian.pool.ntp.org                 .POOL.                             16 p - 64    0    0.0000  0.0000  0.0001
1.debian.pool.ntp.org                 .POOL.                             16 p - 64    0    0.0000  0.0000  0.0001
2.debian.pool.ntp.org                 .POOL.                             16 p - 64    0    0.0000  0.0000  0.0001
3.debian.pool.ntp.org                 .POOL.                             16 p - 64    0    0.0000  0.0000  0.0001
+91.206.8.70                          40.179.132.91                      2 u 33 64 377 23.0846 -17.6747 10.0859
+152.53.44.250                        237.17.284.95                      2 u 36 64 377 23.1949 -18.0053 20.3276
+152.53.15.127                        205.46.178.169                     2 u 35 64 377 31.7621 -19.3559 19.0390
+172.104.154.182                      82.68.30.35                        3 u 31 64 377 35.1608 -16.8842 16.5248
+77.90.40.94                          195.145.119.188                    2 u 38 64 377 32.1426 -19.2371 24.1381
+85.215.166.214                       192.53.103.108                     2 u 45 64 377 45.3741 -15.5881 17.4728
+213.239.234.28                       131.188.3.222                      2 u 33 64 377 33.0521 -17.3490 18.1665
+136.243.7.20                         56.1.129.236                       3 u 37 64 377 41.5441 -0.5694 14.3559
+141.144.246.224                      169.254.169.254                    3 u 34 64 377 37.2749 -18.1869 17.9356
+162.159.200.1                        10.213.8.4                         3 u 37 64 377 24.1801 -17.9570 14.9647
+185.232.69.65                        79.133.44.136                      2 u 19 64 377 35.0523 -17.3967 14.5993
+78.47.168.188                        168.239.11.197                     2 u 27 64 377 38.5162 -20.6345 12.9223
+131.188.3.223                        .PTP.                               1 u 32 64 377 38.6650 -18.4860 12.6227
+81.7.16.52                           131.188.3.222                      2 u 38 64 377 43.9830 -5.8918 8.7890
+185.41.106.152                       192.53.103.108                     2 u 33 64 377 44.8466 -14.1729 12.0271
root@box1-t122:/home/developer# date
Wed Nov 12 10:32:20 UTC 2025
root@box1-t122:/home/developer#

```

6.3.1.9. Active processes

Test ID:	6.3.1.9
Description:	Ensure that the following processes are running: zabbix-proxy, zabbix-agent, telemetry, mqtt-bridge, train-identification-api, mar3, snmpd, nd-redundancy, lldpd, ntp, gpsd-nd
Execution Steps:	Console into CCU Follow steps in first test For each of the processes listed in the description run 'systemctl status <PROCESS>' Each process should report active (running)
Expected Outcome:	The CCU shows the repective processes in a state of active (exited) or in the state of active(running). No process reports 'failed'
Preconditions:	CCU is staged CCU is powered CCU has ethernet wan link or SIMs
Bench:	Devops Bench
Test Method:	Functional testing

Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	
<pre>root@box1-t122:/home/developer# sudo systemctl status zabbix-proxy zabbix-agent telemetry mqtt-bridge train-identification-api mar3 snmpd nd-redundancy lldpd ntp gpsd grep Active: -B2 ● zabbix-proxy.service - Zabbix Proxy (50Lits3) Loaded: loaded (/lib/systemd/system/zabbix-proxy.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:43 UTC; 36min ago Unit telemetry.service could not be found. -- ● zabbix-agent.service - Zabbix Agent Loaded: loaded (/lib/systemd/system/zabbix-agent.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:43 UTC; 36min ago -- ● mqtt-bridge.service - Nomad MQTT Bridge Loaded: loaded (/lib/systemd/system/mqtt-bridge.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:43 UTC; 36min ago -- ● train-identification-api.service - 21Net Train Identification API Loaded: loaded (/lib/systemd/system/train-identification-api.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:43 UTC; 36min ago -- ● mar3.service - 21Net Mobile Access Router Service Loaded: loaded (/lib/systemd/system/mar3.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:43 UTC; 36min ago -- ● snmpd.service - Simple Network Management Protocol (SNMP) Daemon. Loaded: loaded (/lib/systemd/system/snmpd.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:43 UTC; 36min ago -- ● nd-redundancy.service - ND Redundancy Loaded: loaded (/lib/systemd/system/nd-redundancy.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:43 UTC; 35min ago -- ● lldpd.service - LLDP daemon Loaded: loaded (/lib/systemd/system/lldpd.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:43 UTC; 35min ago -- ● ntpsec.service - Network Time Service Loaded: loaded (/lib/systemd/system/ntpsec.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:45 UTC; 35min ago -- ● gpsd-nd.service - GPS (Global Positioning System) Daemon Loaded: loaded (/lib/systemd/system/gpsd-nd.service; enabled; preset: disabled) Active: active (running) since Wed 2025-11-12 10:00:43 UTC; 36min ago</pre>	

6.3.2. OBN Functionality

6.3.2.1. OBN Automatic discovery

Test ID:	6.3.2.1
Description:	Check that following boot, that OBN detects all connected devices to network (that are meant to be seen)
Execution Steps:	Visually Ensure that all devices are connected Connect to CCU via SSH or console Ensure that the CCU has been at least online for ten minutes using the ‘uptime’ command

	Run for following commands to perform an obn discovery and validation <i>'obn discover && obn report && obn validate'</i> After all three commands run a table will be drawn in console Confirm that all devices connected (excluding CCU) are in the table
Expected Outcome:	All switches and APs that are connected are detected by obn and displayed in the table.
Preconditions:	Bench is fully built and has been online for five minutes Previous tests passed
Bench:	Devops Bench
Test Method:	Functional testing
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	

```
root@box1-t122:/home/developer# obn discover ; obn report ; obn validate ; date
+-----+
|                                     Backbone switch overview                                     |
+-----+-----+-----+-----+-----+-----+
| COACH | DEVICE | IP           | FIRMWARE | CONFIG |
+-----+-----+-----+-----+-----+
| 1      | 1      | 10.179.122.185 ✓ | 7.4.2 ✓  | 4t-A1-v2-250 ✓ |
| 1      | 2      | 10.179.122.186 ✓ | 7.4.2 ✓  | 4t-A2-v2-250 ✓ |
| 1      | 3      | 10.179.122.183 ✓ | 7.4.2 ✓  | 4t-A3-v2-250 ✓ |
| 2      | 1      | 10.179.122.187 ✓ | 7.4.2 ✓  | 4t-G1-v2-250 ✓ |
| 2      | 2      | 10.179.122.196 ✓ | 7.4.2 ✓  | 4t-G2-v2-250 ✓ |
| 2      | 3      | 10.179.122.179 ✓ | 7.4.2 ✓  | 4t-G3-v2-250 ✓ |
| 3      | 1      | 10.179.122.180 ✓ | 7.4.2 ✓  | 4t-E1-v2-250 ✓ |
| 3      | 2      | 10.179.122.184 ✓ | 7.4.2 ✓  | 4t-E2-v2-250 ✓ |
| 3      | 3      | 10.179.122.182 ✓ | 7.4.2 ✓  | 4t-E3-v2-250 ✓ |
| 4      | 1      | 10.179.122.188 ✓ | 7.4.2 ✓  | 4t-B1-v2-250 ✓ |
| 4      | 2      | 10.179.122.178 ✓ | 7.4.2 ✓  | 4t-B2-v2-250 ✓ |
| 4      | 3      | 10.179.122.189 ✓ | 7.4.2 ✓  | 4t-B3-v2-250 ✓ |
+-----+-----+-----+-----+-----+

+-----+
|                                     Backbone access point overview                                     |
+-----+-----+-----+-----+-----+
| COACH | DEVICE | IP           | FIRMWARE | CONFIG |
+-----+-----+-----+-----+-----+
| 3      | 1      | 10.179.122.138 ✓ | 6.10.0-0 ✓ | AP1-v1 ✓ |
+-----+-----+-----+-----+-----+

Wed Nov 12 10:37:53 UTC 2025
```

6.3.2.2. OBN detects valid firmware

Test ID:	6.3.2.2
Description:	Ensure that obn detects the firmware of the devices and evaluates whether it is valid or not
Execution Steps	Visually Ensure that all devices are connected Connect to CCU via SSH or console Ensure that the CCU has been online for at least ten minutes using the 'uptime' command. Run for following commands to perform an obn discovery and validation <i>'obn discover && obn report && obn validate'</i> After all three commands run, a table will be displayed. Confirm that all devices connected (excluding CCU) are in the table In the firmware columns, confirm the versions displayed matches those in section 4.1 or the hardware release note.

	Firmware can be different (requires upgrade) but cannot be 'unknown'.
Expected Outcome:	OBN states the device firmware versions and whether the firmware of the devices is valid.
Preconditions:	Bench system is fully built and has been online for five minutes Previous tests passed
Bench:	Devops Bench
Test Method:	Functional testing
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence: Check previous test	

6.3.2.3. OBN Successful update of device configuration

Test ID:	6.3.2.3
Description:	Test the ability to update device configurations. Devices (both switches and AP's) should be able to have their configuration updated.
Execution Steps:	Ensure CCU has been booted for at least five minutes This allows obn to auto-discover all devices Connect to the CCU via ssh or console Run obn validate to confirm all devices present Select the IP of the switch or AP you wish to update In a separate terminal connected to the CCU begin a ping to that IP

	In the main window run the command substituting <IP> for the IP of the device you have chosen 'obn update c <IP>' The obn update runs without error. After a few seconds the device should reboot, and pings will stop Wait five minutes Run 'obn discover && obn report && obn validate' Check if the device is present (potentially under a different IP)
Expected Outcome:	OBN can be called to update a device configuration When updating, the device will be rebooted and not reachable. It may come back up under a new IP
Preconditions:	Bench is fully built All APs and Switches connected to the bench are confirmed to be reachable by completing test 6.3.2.1 Previous tests passed
Bench:	Devops Bench
Test Method:	Functional testing
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	

```
root@box1-t122:/home/developer# obn validate
```

Backbone switch overview					
COACH	DEVICE	IP	FIRMWARE	CONFIG	
1	1	10.179.122.185 ✓	7.4.2 ✓	4t-A1-v2-250	(nv4-A1-v5) ✗
1	2	10.179.122.192 ✓	7.4.2 ✓	4t-A2-v2-250	(nv4-A2-v5) ✗
1	3	10.179.122.183 ✓	7.4.2 ✓	4t-A3-v2-250	(nv4-A3-v5) ✗
2	1	10.179.122.190 ✓	7.4.2 ✓	4t-G1-v2-250	(nv4-G1-v5) ✗
2	2	10.179.122.197 ✓	7.4.2 ✓	4t-G2-v2-250	(nv4-G2-v5) ✗
2	3	10.179.122.191 ✓	7.4.2 ✓	4t-G3-v2-250	(nv4-G3-v5) ✗
3	1	10.179.122.181 ✓	7.4.2 ✓	4t-E1-v2-250	(nv4-E1-v5) ✗
3	2	10.179.122.198 ✓	7.4.2 ✓	4t-E2-v2-250	(nv4-E2-v5) ✗
3	3	10.179.122.193 ✓	7.4.2 ✓	4t-E3-v2-250	(nv4-E3-v5) ✗
4	1	10.179.122.194 ✓	7.4.2 ✓	4t-B1-v2-250	(nv4-B1-v5) ✗
4	2	10.179.122.195 ✓	7.4.2 ✓	4t-B2-v2-250	(nv4-B2-v5) ✗
4	3	10.179.122.189 ✓	7.4.2 ✓	4t-B3-v2-250	(nv4-B3-v5) ✗

Backbone access point overview				
COACH	DEVICE	IP	FIRMWARE	CONFIG
3	1	10.179.122.138 ✓	6.10.0-0 (6.11.2-0) ✗	AP1-v1 (v1-AP1) ✗

```

root@box1-t122:/home/developer# obn update c 10.179.122.190
[+] creating TFTP firmware folder
[+] 10.179.122.190: updating configuration on SW with mac a0:59:3a:d0:60:e0
[+] 10.179.122.190: starting configuration update
[+] 10.179.122.190: configuration update applied, device rebooting.
[+] waiting for 5 minutes
root@box1-t122:/home/developer# obn discover ; obn r
report rules
root@box1-t122:/home/developer# obn discover ; obn report ; obn validate

```

Backbone switch overview					
COACH	DEVICE	IP	FIRMWARE	CONFIG	
1	1	10.179.122.185 ✓	7.4.2 ✓	4t-A1-v2-250 (nv4-A1-v5) ✗	
1	2	10.179.122.192 ✓	7.4.2 ✓	4t-A2-v2-250 (nv4-A2-v5) ✗	
1	3	10.179.122.183 ✓	7.4.2 ✓	4t-A3-v2-250 (nv4-A3-v5) ✗	
2	1	10.179.122.190 ✓	7.4.2 ✓	nv4-G1-v5-122 ✓	
2	2	10.179.122.197 ✓	7.4.2 ✓	4t-G2-v2-250 (nv4-G2-v5) ✗	
2	3	10.179.122.191 ✓	7.4.2 ✓	4t-G3-v2-250 (nv4-G3-v5) ✗	
3	1	10.179.122.181 ✓	7.4.2 ✓	4t-E1-v2-250 (nv4-E1-v5) ✗	
3	2	10.179.122.198 ✓	7.4.2 ✓	4t-E2-v2-250 (nv4-E2-v5) ✗	
3	3	10.179.122.193 ✓	7.4.2 ✓	4t-E3-v2-250 (nv4-E3-v5) ✗	
4	1	10.179.122.194 ✓	7.4.2 ✓	4t-B1-v2-250 (nv4-B1-v5) ✗	
4	2	10.179.122.195 ✓	7.4.2 ✓	4t-B2-v2-250 (nv4-B2-v5) ✗	
4	3	10.179.122.189 ✓	7.4.2 ✓	4t-B3-v2-250 (nv4-B3-v5) ✗	

Backbone access point overview				
COACH	DEVICE	IP	FIRMWARE	CONFIG
3	1	10.179.122.138 ✓	6.10.0-0 (6.11.2-0) ✗	AP1-v1 (v1-AP1) ✗

Test passed: Switch G1

Backbone access point overview				
COACH	DEVICE	IP	FIRMWARE	CONFIG
3	1	10.179.122.138 ✓	6.10.0-0 (6.11.2-0) ✗	AP1-v1 (v1-AP1) ✗

Backbone access point overview				
COACH	DEVICE	IP	FIRMWARE	CONFIG
3	1	10.179.122.138 ✓	6.10.0-0 (6.11.2-0) ✗	RT610LV-dosto-00145a048048-v1-AP1 ✓

Test passed: AP1

6.3.2.4. OBN Successful update of device firmware

Test ID:	6.3.2.4
Description:	Test the ability to update device firmware versions. Devices (both switches and APs) should be able to have their firmware updated.
Execution Steps:	Ensure CCU has been booted for at least five minutes This allows obn to auto-discover all devices Connect to the CCU via ssh or console Run obn validate to confirm all devices present Select the IP of the switch or AP you wish to update In a separate terminal connected to the CCU begin a ping to that IP In the main window run the command substituting <IP> for the IP of the device you have chosen 'obn update f <IP>' The obn update runs without error. After a few seconds the device should reboot and pings will stop Wait five minutes Run 'obn discover && obn report && obn validate' Check if the device is present (potentially under a different IP)
Expected Outcome:	OBN can be called to update a device, when updating the device will be rebooted
Preconditions:	Bench is fully built and has been online for five minutes Previous tests passed
Bench:	Devops Bench

Test Method:	Functional testing
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence: AP FW ran, but no change SW is already up to date	

6.3.3. Wan Functionality

6.3.3.1. Modem Detection and Restarting

Test ID:	6.3.3.1
Description:	Ensure all Modems are detected in the CCU, all modems connect to base station and can be restarted through the marcli commands
Execution Steps:	Log into the CCU via console Escalate privileges to root using sudo su Run command lsusb Confirm that four Telit Modems appear in the results Run 'marcli all' All two modems should appear in state 'CONNECTED' Next test that all modems can be restarted In the terminal run 'marcli modem all powercycle' All modems should disconnect and be reset Use marcli to verify that they reset and return to the active state.
Expected Outcome:	All modems appear in state ACTIVE All Modems found in lsusb All modems can be reset
Preconditions:	All modems installed in CCU, SIMS also installed. CCU online for at least five minutes
Bench:	Devops Bench
Test Method:	Functional
Test Device:	Linux

Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	
<pre> root@box1-t122:/home/developer# marcli modem ce0p0 powercycle Command executed successfully. root@box1-t122:/home/developer# marcli modem state powered geo-disabled sim operator cell lac network band iccid ----- ce0p0: UNKNOWN TRUE FALSE 1 0 0 0 LTE B3 89320272200211775960 ce1p0: CONNECTED TRUE FALSE 4 23203 10028555 14061 LTE B3 89430103122049054928 ce2p0: CONNECTED TRUE FALSE 1 23201 2063629 17006 LTE B3 89430103122049054928 ce3p0: CONNECTED TRUE FALSE 4 23203 10028555 14061 LTE B3 89320272200211778690 </pre>	

6.3.3.2. Modem availability and expected operators

Test ID:	6.3.3.2
Description:	All modems are detected by the CCU and it detects the correct providers of the SIM cards installed.
Execution Steps:	Get the expected providers from the common.yaml Run 'marcli modem all' Check that the APNs/providers specified in the common.yaml match the results given in the network section
Expected Outcome:	All modems are detected by the CCU, each SIM card is picked up by the CCU and data passes in respective country (UK for this test). The providers on the CCU match those specified in the common.yaml
Preconditions:	Modems are all installed, and SIMs are present. CCU has been online for at least 10 minutes
Bench:	Devops Bench
Test Method:	Functional
Test Device:	CCU
Execution Date:	12/11/2025

Executer:	Davud Zejnelovic
Evidence:	
<pre> root@box1-t122:/home/developer# marcli modem state powered geo-disabled sim operator cell lac network band iccid ----- ce0p0: UNKNOWN TRUE FALSE 1 0 0 0 LTE B3 89320272200211775960 ce1p0: CONNECTED TRUE FALSE 4 23203 10028555 14061 LTE B3 89430103122040054928 ce2p0: CONNECTED TRUE FALSE 1 23201 2063629 17006 LTE B3 89320272200211778690 ce3p0: CONNECTED TRUE FALSE 4 23203 10028555 14061 LTE B3 89320272200211778690 root@box1-t122:/home/developer# █ </pre>	
No GPS in Lab -> Default loaded correctly -> Sim switching observed on boot.	

6.3.3.3. Link Aggregation and TCP Handoff of multiple streams

Test ID:	6.3.3.3
Description:	Validate that TCP traffic is aggregated across all available modems, and that TCP traffic is handed off to another modem upon failure of an individual modem
Execution Steps:	1) Run <code>{saturate.sh}</code> 2) Check running <code>{watch marcli all}</code> and validate that download is being aggregated across all modems. 3) Kill one modem via <code>{marcli modem ce*p* poweroff}</code> command and validate that current download streams are handed off to remaining modem 4) Kill remaining modems one at a time <code>{marcli modem ce*p* poweroff}</code> until only one modem remains, validating that TCP handoff of streams occurs upon each modem being disabled. 5) Power modems back on one at a time <code>{marcli modem ce*p* poweron}</code> ensuring that some of the download streams are handed back off to recovered modems.
Expected Outcome:	When a modem is lost, all traffic going through it is switched to a different available modem
Preconditions:	CCU is online, tunnel is up

	Should be completed in conjunction with section 6.3.8
Bench:	Devops
Test Method:	Functional/ Robustness testing
Test Device:	CCU
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic

Evidence:

```
Last login: Wed Nov 12 10:04:00 2025 from 192.168.69.130
developer@box1-t122:~$ sudo su
root@box1-t122:/home/developer# saturate.sh
```

We can see it distributes between the 3 active modems.

```
^Croot@box1-t122:/home/developer# marcli all
MAR INFORMATION:
*****
ce0p0: st: DOWN bs: 0 br: 0 al: -1 at: 0.000 Mbit/s au: 0.000 Mbit/s ap: 0% ns: 0 tps: 0 ha: IPv4
ce1p0: st: ACTIVE bs: 6628603 br: 181822036 al: 120 at: 24.005 Mbit/s au: 0.520 Mbit/s ap: 0% ns: 31 tps: 0 ha: IPv4
ce2p0: st: ACTIVE bs: 26067032 br: 380643999 al: 217 at: 157.980 Mbit/s au: 1.711 Mbit/s ap: 0% ns: 80 tps: 0 ha: IPv4
ce3p0: st: ACTIVE bs: 5862842 br: 132544857 al: 306 at: 285.904 Mbit/s au: 6.667 Mbit/s ap: 0% ns: 35 tps: 0 ha: IPv4
lan2: st: DOWN bs: 0 br: 0 al: -1 at: 0.000 Mbit/s au: 0.000 Mbit/s ap: 0% ns: 0 tps: 0 ha: IPv4
wlan0: st: DOWN bs: 0 br: 0 al: -1 at: 0.000 Mbit/s au: 0.000 Mbit/s ap: 0% ns: 0 tps: 0 ha: IPv4

total: bs: 38558477 br: 695010892 at: 467.888 Mbit au: 8.898 Mbit/s

HA (IPv4): 77.237.62.210
HA (IPv6): None

MODEM INFORMATION:
*****
state powered geo-disabled sim operator cell lac network band iccid
---- -
ce0p0: UNKNOWN TRUE FALSE 1 0 0 0 LTE B3 89320272200211775960
ce1p0: CONNECTED TRUE FALSE 4 23203 10028555 14061 LTE B3 89430103122049054928
ce2p0: CONNECTED TRUE FALSE 1 23201 2063629 17006 LTE B3 89430103122049054928
ce3p0: CONNECTED TRUE FALSE 4 23203 10028545 14061 LTE B7 89320272200211778690

GEO INFORMATION:
*****
GPS updates enabled: TRUE
Last update: Wed Nov 12 10:49:28 UTC 2025
Valid: FALSE
Latitude: -1.00000
Longitude: -1.00000
Altitude: -1.00000
Heading: -1.00
Speed: 0 km/h
```

```

root@box1-t122:/home/developer# marcli all
MAR INFORMATION:
*****
ce0p0: st: DOWN bs: 0 br: 0 al: -1 at: 0.000 Mbit/s au: 0.000 Mbit/s ap: 0% ns: 0 tps: 0 ha: IPv4
celp0: st: ACTIVE bs: 18078898 br: 723247967 al: 123 at: 3.478 Mbit/s au: 0.082 Mbit/s ap: 0% ns: 207 tps: 0 ha: IPv4
ce2p0: st: ACTIVE bs: 31963790 br: 712979412 al: 37 at: 1.945 Mbit/s au: 0.086 Mbit/s ap: 0% ns: 159 tps: 0 ha: IPv4
ce3p0: st: ACTIVE bs: 12621542 br: 433248964 al: 123 at: 0.103 Mbit/s au: 0.010 Mbit/s ap: 0% ns: 85 tps: 0 ha: IPv4
lan2: st: DOWN bs: 0 br: 0 al: -1 at: 0.000 Mbit/s au: 0.000 Mbit/s ap: 0% ns: 0 tps: 0 ha: IPv4
wlan0: st: DOWN bs: 0 br: 0 al: -1 at: 0.000 Mbit/s au: 0.000 Mbit/s ap: 0% ns: 0 tps: 0 ha: IPv4

total: bs: 62664230 br: 1869476343 at: 5.525 Mbit au: 0.178 Mbit/s

HA (IPv4): 77.237.62.210
HA (IPv6): None

MODEM INFORMATION:
*****
state powered geo-disabled sim operator cell lac network band iccid
----
ce0p0: ENABLED TRUE FALSE 1 0 0 0 0 89430103122049054829
celp0: CONNECTED TRUE FALSE 4 23203 10028555 14061 LTE B3 89320272200211775960
ce2p0: CONNECTED TRUE FALSE 1 23201 2063629 17006 LTE B3 89430103122049054928
ce3p0: CONNECTED TRUE FALSE 4 23203 10028545 14061 LTE B7 89320272200211778690

GEO INFORMATION:
*****
GPS updates enabled: TRUE
Last update: Wed Nov 12 10:49:56 UTC 2025

Valid: FALSE
Latitude: -1.00000
Longitude: -1.00000
Altitude: -1.00000
Heading: -1.00
Speed: 0 km/h

```

After killing the script

6.3.3.4. Modem Testing and Validation

Test ID:	6.3.3.4
Description:	Test that client device TCP traffic is seamlessly handed off to another modem without interruption to the end user when the modem their TCP stream is going through fails.
Execution Steps:	Actions in blue are to perform on a client device Actions in black are to performed on the device used to access the CCU 1. Connect a client device to the network using the Staff or passenger SSID <i>Follow the steps outline in 6.3.8 to connect the device to the passenger network</i> 2. Begin running a ping to 8.8.8.8 from the client device 3. Once the device is connected and ping is running connect to the CCU with a separate device

	4. Run cp-list.sh to get the IP of the device 5. Next run marcli session and look for the device IP under each modem 6. Once you have found the IP note the modem for the future steps 7. Run command marcli modem <modem> reset 8. The ping on the client device should continue (one packet may be lost) 9. Check on the CCU that the ping has moved to a different modem by running marcli session and look for the client device IP 10. The IP should be under a different modem
Expected Outcome:	When a modem is lost, all traffic going through it is switched to a different available modem
Preconditions:	CCU is online, tunnel is up Should be completed in conjunction with section 6.3.8
Bench:	Devops
Test Method:	Functional/ Robustness testing
Test Device:	CCU / Android
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	

```

ce2p0
id: 1395, src: 10.179.122.254, dst: 68.210.112.58, prt: udp, ts: 0 kbps, tr: 0 kbps, bs: 676728, br: 0, st: ESTABLISHED
id: 4347, src: 10.179.122.254, dst: 172.16.41.81, prt: tcp, ts: 0 kbps, tr: 0 kbps, bs: 14752865, br: 1481650, st: ESTABLISHED
id: 16796, src: 10.179.122.254, dst: 157.240.0.61, prt: tcp, ts: 0 kbps, tr: 0 kbps, bs: 2984, br: 3223, st: ESTABLISHED
id: 17263, src: 192.168.69.130, dst: 10.179.122.1, prt: tcp, ts: 0 kbps, tr: 0 kbps, bs: 37076, br: 11832, st: ESTABLISHED
id: 17322, src: 192.168.208.11, dst: 140.153.240.65, prt: tcp, ts: 0 kbps, tr: 0 kbps, bs: 468, br: 0, st: ESTABLISHED
id: 17377, src: 10.179.122.129, dst: 172.19.5.1, prt: tcp, ts: 0 kbps, tr: 0 kbps, bs: 360, br: 0, st: ESTABLISHED

root@box1-t122:/home/developer# marcli modem ce2p0 poweroff
Command executed successfully.
root@box1-t122:/home/developer# marcli modem
state      powered      geo-disabled  sim      operator      cell      lac      network      band      iccid
-----
ce0p0:  ENABLED      TRUE         FALSE      1          0          0          0          -----
ce1p0:  CONNECTED   TRUE         FALSE      4          23203      10020555  14061      LTE      B3      89320272200211775960
ce2p0:  CONNECTED   FALSE        FALSE      1          23201      2063629   17006      LTE      B3      89430103122049054928
ce3p0:  CONNECTED   TRUE         FALSE      4          23203      10020555  14061      LTE      B3      89320272200211776690

root@box1-t122:/home/developer# marcli session | grep 192.168.208
root@box1-t122:/home/developer# marcli session
ce3p0
id: 1395, src: 10.179.122.254, dst: 68.210.112.58, prt: udp, ts: 0 kbps, tr: 0 kbps, bs: 711360, br: 0, st: ESTABLISHED
id: 4347, src: 10.179.122.254, dst: 172.16.41.81, prt: tcp, ts: 0 kbps, tr: 0 kbps, bs: 15540556, br: 1559122, st: ESTABLISHED
id: 17077, src: 10.179.122.254, dst: 157.240.0.61, prt: tcp, ts: 0 kbps, tr: 0 kbps, bs: 653, br: 663, st: ESTABLISHED
id: 17097, src: 192.168.69.130, dst: 10.179.122.1, prt: tcp, ts: 0 kbps, tr: 0 kbps, bs: 43820, br: 20904, st: ESTABLISHED
id: 18043, src: 10.179.122.129, dst: 172.19.5.1, prt: tcp, ts: 0 kbps, tr: 0 kbps, bs: 360, br: 0, st: ESTABLISHED

```

No delay on the Android ping test

6.3.4. On-board functionality

6.3.4.1. Switch Functionality

Test ID:	6.3.4.1
Description:	Check that switches are physically connected correctly, and that status lights are visible.
Execution Steps:	1. Setup bench as shown in technical diagram Referenced above 2. Ensure that all cables are properly situated and connected 3. Check status lights on each switch and ensure they meet the expected outcomes below
Expected Outcome:	Switch port lights on main switch: - Port 1 is lit and green - Port 2 is lit and green - Port 5 is lit, green and poe is lit green. Switch port lights on secondary switch: - Port 1 is lit and green - Port 5 is lit, green and poe is lit green.
Preconditions:	All previous tests passed
Bench:	Devops Bench
Test Method:	Functional
Test Device:	Linux
Execution Date:	17/09/2025
Executer:	Piotr Romanowski
Evidence:	
Pass	

6.3.4.2. AP Functionality

Test ID:	6.3.4.2
----------	---------

Description:	Check that APs are physically connected correctly, and relevant port status lights are visible and green.
Execution Steps:	1. Setup bench as shown in technical diagram Referenced above 2. Ensure that all cables are properly situated and connected 3. Ensure that 1000M link is green. 4. That fail light is not orange/red. May have hint of green from adjoining light 5. That the AP passenger SSIDs are broadcasting
Expected Outcome:	APs are online, gigabit link light active Passenger SSIDs are broadcasting No orange/red lights reported on APs
Preconditions:	6.3.4.1 passed - Switch Functionality
Bench:	Devops Bench
Test Method:	Functional
Test Device:	Linux
Execution Date:	17/09/2025
Executer:	Piotr Romanowski
Evidence: Pass	

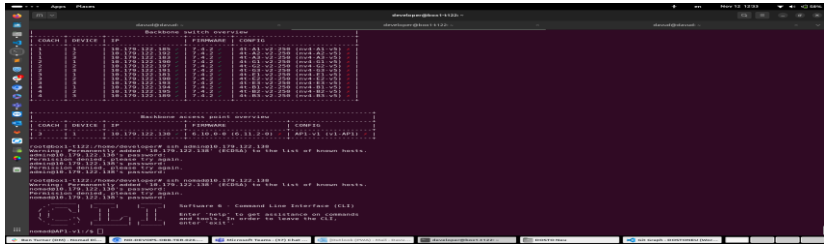
6.3.4.3. DHCP and User Session Counting

Test ID:	6.3.4.3
Description:	Test to see that passengers are allocated an IP address when they connect. Ensure that the Passenger can be counted on the CCU and that session details are available

Execution Steps:	1. Connect a device to the passenger network 2. Ensure that it is prompted to sign-in 3. Sign-in to Wi-fi 4. Connect to CCU using secondary device via SSH and console 4. Run command cp-stat.sh 5. Check that device IP reported for the device matches the device listed in cp-stat.sh
Expected Outcome:	CCU reports valid information about the number of users connected Connected device is visible in cp-stat.sh
Preconditions:	Test 6.3.4.3 passed Should be executed in conjunction with test 6.3.8.1
Bench:	Devops bench
Test Method:	Functional
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence: Works with Android phone (Huawei). OBB Landing page comes up and requires a checkbox and then internet access was provided for the test device, see modem session test.	

6.3.4.4. AP Accessibility

Test ID:	6.3.4.4
Description:	Check that APs can be accessed via SSH from the CCU
Execution Steps:	Access CCU via SSH or console

	Run <code>obn validate</code> or <code>arp -n</code> and look for a AP IP <i>IP should be 10.167.X.[220-229]</i> On the CCU run <code>ssh@<IP></code> Enter the password for the AP You should now have access to the AP
Expected Outcome:	APs are accessible via SSH through the CCU
Preconditions:	6.3.4.1 passed - Switch Functionality 6.3.4.2 passed - AP Functionality
Bench:	Devops Bench
Test Method:	Functional
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	
	

6.3.4.5. Switch Accessibility

Test ID:	6.3.4.5
Description:	Check that switches can be accessed from the CCU
Execution Steps:	Access CCU via SSH or console

	Run <code>obn validate</code> or <code>arp -n</code> and look for a AP IP <i>IP should be 10.167.X.[220-229]</i> On the CCU run <code>ssh@<IP></code> Enter the password for the AP You should now have access to the AP
Expected Outcome:	Switches are accessible through SSH when connected to the CCU
Preconditions:	6.3.4.1 passed - Switch Functionality
Bench:	Devops Bench
Test Method:	Functional
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence: <pre> root@box1-t112:/home/developer# ssh admin@10.178.112.184 Warning: Permanently added '10.178.112.184' (ED25519) to the list of known hosts. (admin@10.178.112.184) Password: Last login: Thu Jan 1 21:47:23 1970 from 10.178.112.129 ** Konuendo Platform Configuration Shell ** ** Railway Switch Version ** ** Press '?' for Help ** A@dosto-T1-Sw600-B-3> </pre>	

6.3.5. CCU API

6.3.5.1. Ensure GPS API calls to localhost are possible

Test ID	6.3.5.1
Description	Ensure that the CCU GPS API feed is available on the CCU and can be accessed through CCU
Execution steps:	Access the CCU through Console or SSH Once on the terminal run command 'curl localhost:1816/gps/data' Output should state success:true along with similar co-ordinates to that of the marcli geo run in pre-conditions
Expected Outcome	On curling the CCU a json format API result is returned containing a success message
Preconditions	Test 6.3.1.7 (tunnel form test) passed, tunnel connected This test should be executed in conjunction or after test 6.3.7.1 GPS cable connected CCU online for five minutes GPS confirmed with 'marcli geo'
Test Type	Devops bench
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	
<pre>root@box1-t121:/home/developer# curl http://localhost:1816/gps/data {"success":false}root@box1-t121:/home/developer#</pre>	

No GPS in Lab -> But is expected to work as it was already tested

6.3.5.2. Ensure Train Identification API calls to localhost are possible

Test ID	6.3.5.2
Description	Ensure that the CCU train identification API feed is available on the CCU and can be accessed through CCU
Execution steps:	Access the CCU through Console or SSH Once on the terminal run command 'curl localhost:1817/identification' Output should state the train ID along with its project (39) and its external ID if it has one defined by NMS
Expected Outcome	On curling the CCU a json format API result is returned containing a success message
Preconditions	Test 6.3.1.7 (tunnel form test) passed, tunnel connected CCU online for five minutes
Test Type	Devops bench
Test Device:	Linux
Execution Date:	12/11/2025
Executer:	Davud Zejnelovic
Evidence:	
<pre>root@box1-t122:/home/developer# curl localhost:1817/identification {"projectId":50,"trainId":4122,"boxId":1,"externalId":"Dostoneu_Bench_0BB_4C"}root@box1-t122:/home/developer#</pre>	

6.3.5.3. Ensure Train captive portal API calls to localhost are possible

Test ID	6.3.5.3
Description	Ensure that the CCU train captive portal API feed is available on the CCU and can be accessed through CCU
Execution steps:	Access the CCU through Console or SSH Once on the terminal run command 'curl localhost:1820/list Output should state 'list succeeded' If any passenger devices are connected it will also contain information on them
Expected Outcome	On curling the CCU a json format API result is returned containing a success message
Preconditions	Test 6.3.1.7 (tunnel form test) passed, tunnel connected CCU online for five minutes A Device connected to the passenger wi-fi (optional)
Test Type	Devops bench
Test Device:	Linux
Execution Date:	22/09/2025
Executer:	Piotr Romanowski
Evidence:	
<pre>root@box1-t121:/home/developer# curl localhost:1820/list {"grants":[],"status_code":200,"status_description":"list succeeded"}root@box1-t121:/home/developer#</pre>	

6.3.5.4. Ensure Engineering Pages are available

Test ID	6.3.5.4
---------	---------

Description	Ensure that the CCU Engineering Pages are available, and data is aCCurate
Execution steps:	In web browser, navigate to: *CCU IP address*:8005 Login with: username: stanley password: waffles
Expected Outcome	Engineering pages are available, and data provided is aCCurate
Preconditions	CCU online System updated
Test Type	Devops bench
Test Device:	Linux
Execution Date:	22/09/2025
Executer:	Szymon Dutkiewicz / Piotr Romanowski

Evidence:



Engineering Pages - Train identification

Key	Value
Project ID	50
Train ID	112
Unit ID	1
Hostname	box1-t112.dostoneu-bench.21net.com

engineering.nomadrail.com8005

Not securehttps://engineering.nomadrail.com:8005

NOMAD DIGITAL U...UDEMY AWSMail - Szymon Dutki...DEVOPS MainIP command lin...BigQuery - OandM...

Incognito (2)

Sign in

https://engineering.nomadrail.com:8005

Username

Password

Sign inCancel

18/09/2025

```
root@box1-t121:/home/developer# curl -s -k -u oebbStaff:W1f1@100Mph -L "https://localhost:8005" | grep -Eo "Nomad Digital Engineering Pages"
Nomad Digital Engineering Pages
root@box1-t121:/home/developer#
```

18/09/2025

```
root@box1-t121:/home/developer# curl -s -k -u oebbStaff:W1f1@100Mph -L "https://localhost:8005" | grep -Eo "Nomad Digital Engineering Pages"
Nomad Digital Engineering Pages
root@box1-t121:/home/developer#
```

6.3.6. System Update

6.3.6.1. System Update

Test ID	6.3.6.1
Description	Ensure that the system update feature runs and makes the required changes to the system. Reboot system and ensure that these changes take effect
Execution Steps	<i>Execution steps rely on obn being present on the system, should obn not be present use '21net-security' instead</i> Connect to CCU via console or ssh Escalate privileges to root Run 'nd- systemupdate.sh shell' Type 'apt purge nd-obn' Type 'exit 0' This is to make a manual change to the system, removing obn Reboot the system Once the system has rebooted log back in The obn command should no longer work Run 'nd-systemupdate.sh up' The CCU should then begin to override the manual change Reboot the CCU Log back in Obj should now be installed and the command should work again
Expected Outcome	Following a manual change being made to the system (and rebooted). When the system update is

	run the manual change is removed and on reboot the manual change is not in the running system
Preconditions	Basic System functionality tests passed
Test Type	Devops bench
Test Device	Linux
Execution Date	12/11/2025
Executer	Davud Zejnelovic
Evidence:	
<pre> Transactional update started Options: up ===== Remote version: 7ce1d29ce65f451a92d5bdd51155d4fdaded304c Local version: 28f51d9e68615c7ea22450d398da13a2c126d0d0 Remote tag: - Local tag: - Remote environment: dostoneu_migration_mar5 Local environment: dostoneu_migration_mar5 NomadConnect remote ref: a33b27902c4aeb1d983d99eaac3b93447f1ealb7 NomadConnect local ref: a33b27902c4aeb1d983d99eaac3b93447f1ealb7 NomadConnect remote tag: 2025.2.1 NomadConnect local tag: 2025.2.1 ===== Update required, starting update... Certificate will not expire Info: Using environment 'dostoneu_migration_mar5' Info: Retrieving pluginfacts Info: Retrieving plugin Info: Loading facts Info: Caching catalog for box1-t122.dostoneu-bench.21net.com Info: Applying configuration version '7ce1d29ce65f451a92d5bdd51155d4fdaded304c' Notice: /Stage[main]/Base_profile::Puppet::Mcollective/Service[mcollective]/ensure: ensure changed 'running' to 'stopped' (corrective) Notice: /Stage[main]/Apt::Update/Exec[apt update]/returns: executed successfully (corrective) Notice: /Stage[main]/Base_profile::Ccu/Exec[apt-update]/returns: Notice: /Stage[main]/Base_profile::Ccu/Exec[apt-update]/returns: WARNING: apt does not have a stable CLI interface. Use with caution in scripts. Notice: /Stage[main]/Base_profile::Ccu/Exec[apt-update]/returns: Notice: /Stage[main]/Base_profile::Ccu/Exec[apt-update]/returns: Ign:1 http://vmrepo01.ovh2.21net.com/bookworm bookworm InRelease Notice: /Stage[main]/Base_profile::Ccu/Exec[apt-update]/returns: Hit:2 http://vmmirror01.central.21net.com/debian 20250311 bookworm InRelease Notice: /Stage[main]/Base_profile::Ccu/Exec[apt-update]/returns: Hit:3 http://vmrepo01.ovh2.21net.com/bookworm bookworm Release Notice: /Stage[main]/Base_profile::Ccu/Exec[apt-update]/returns: Hit:4 http://vmmirror01.central.21net.com/debian-security 20250311 bookworm-security/InRelease Notice: /Stage[main]/Booker::Ipservice/Service[no-booker-networking.service]: triggered refresh from IP Notice: /Stage[main]/Security::Package/Package[21net-security-dostoneu]/ensure: created (corrective) Notice: /Stage[main]/Base_system::Config/Sysctl[net.ipv4.ping_group_range=5000/Exec[enforce-sysctl-value-not]]: triggered refresh from IP root@box1-t122:/home/developer# dpkg -l grep secu ii 21net-security 1.0.3 all Nomad Connect Security Package ii 21net-security-dostoneu 0.0.8 all Nomad Digital DOSTONEU Security Package ii 21net-security-luna 0.0.12 all Nomad Digital OEBB LUNA Security Package root@box1-t122:/home/developer# dpkg -l grep obn ii nd-obn 2.2.23 all On-board Networking - (c) Nomad Digital ii nd-obn-template-dostoneu-nv4 0.0.2 all Nomad Digital OBN Templates for Dosto Neu train type nv4 root@box1-t122:/home/developer# </pre>	

6.3.6.2. Reverting System Update

Test ID	6.3.6.2
Description	Ensure that when an update places the system into a non-functioning state (I.e. unit will not connect) Reboot system and ensure that these changes take effect

Execution Steps	Follow the steps of 6.3.6.1 Power down the system Remove all SIMS (or remove wan blades) and lan connections Once the system has powered back up console in The unit should be unable to form a tunnel Make an empty manual change to the CCU using 'nd-sytemupdate.sh shell' Followed by 'exit 0' Confirm the empty change with 'cat /.snapshot/state' The VERSION should look like the following: VERSION=07a4feb16440f63368b0594e57e1337f3d93737f_manual_202407021216 <i>Note: the hash might change but the key word is MANUAL</i> Reboot the CCU The CCU will boot into our manual change Wait fifteen minutes The CCU should then begin to override the manual change The CCU should then reboot The VERSION in /.snapshot/state should now just be a commit hash, no MANUAL
Expected Outcome	Following a manual change being made to the system (and rebooted). When the system update is run the manual change is removed and on reboot the manual change is not in the running system
Preconditions	Basic System functionality tests passed
Test Type	Devops bench
Test Device	Linux
Execution Date	12/11/2025
Executer	Davud Zejnelovic

Evidence:

Passed

6.3.7. GPS Functionality

6.3.7.1. GPS Module Operational Testing

Test ID:	6.3.7.1
Description:	Ensure that the CCUs GPS module is operational and that when queried by the GPS, it produces valid GEO coordinates.
Execution steps	Ensure GPS cable is firmly connected Connect to CCU via console or ssh Run 'marcli geo' CCU should produce gps co-ordinates relevant to location If GPS is unavailable check cable, re-position cable
Expected Outcome:	GPS is active and shows GPS information. The GPS co-ordinates produced should corollate to current given position
Preconditions:	CCU online, GPS module and cable installed into CCU GPS cable has good sky visibility
Bench:	Devops bench
Test Method:	Functional
Test Device	Linux
Execution Date	12/11/2025
Executer	Davud Zejnelovic
Evidence: No GPS antenna connected to the bench	

```
root@box1-t121:/home/developer# marcli geo
P GPS updates enabled: TRUE
P Last update: Fri Sep 19 13:26:24 UTC 2025
P Valid: FALSE
P Latitude: -1.00000
P Longitude: -1.00000
u Altitude: -1.00000
u Heading: -1.00
u Speed: 0 km/h
root@box1-t121:/home/developer#
```

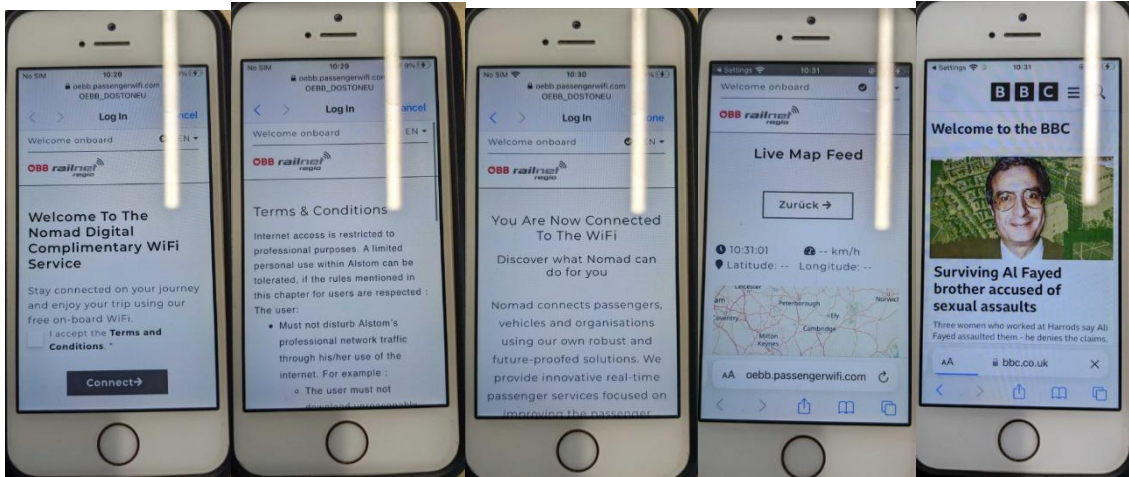

6.3.8. Client/passenger Functionality

When conducting these tests, ensure that between tests the CCU is rebooted to prevent previous tests from influencing future testing.

It also recommended that between tests to have the device forget the Wi-Fi network

6.3.8.1. Passenger Connectivity Flow Apple Mobile

Test ID:	6.3.8.1
Description:	This test looks to follow the expected flow of an Apple client device to ensure all necessary prompts are triggered and displayed when using a mobile client
Execution Steps:	The following outlines the passenger flow for connecting with iPhone 1. Passenger opens Wi-Fi selection 2. Passenger selects *PROJECT_NAME* wifi 3. A prompt to sign in appears within the space of seconds 4. User clicks the prompt and a can browser opens (or browser auto opens) 5. The landing page opens and asks the user to select settings 6. The user then clicks the x in the top corner of the page 7. The user then selects the connect option that appears 8. The user then signs in 9. The user should be taken to a success page (non-existent) 10. The CNA browser will then close (user may have to press) 11. The user can then navigate to ifconfig.io or bbc.com or ping an external IP address 12. Depending on the website, the user is presented with the IP of the gateway or the bbc.co.uk website. Or users ping attempts are successful
Expected Outcome:	This test is passed when the client device can follow all steps given The client is presented with internet access on completion of the above steps
Preconditions:	System configured All APs connected are correctly configured Mar tunnel is up and is allowing connectivity
Bench:	Devops bench
Test Method:	Functional Test

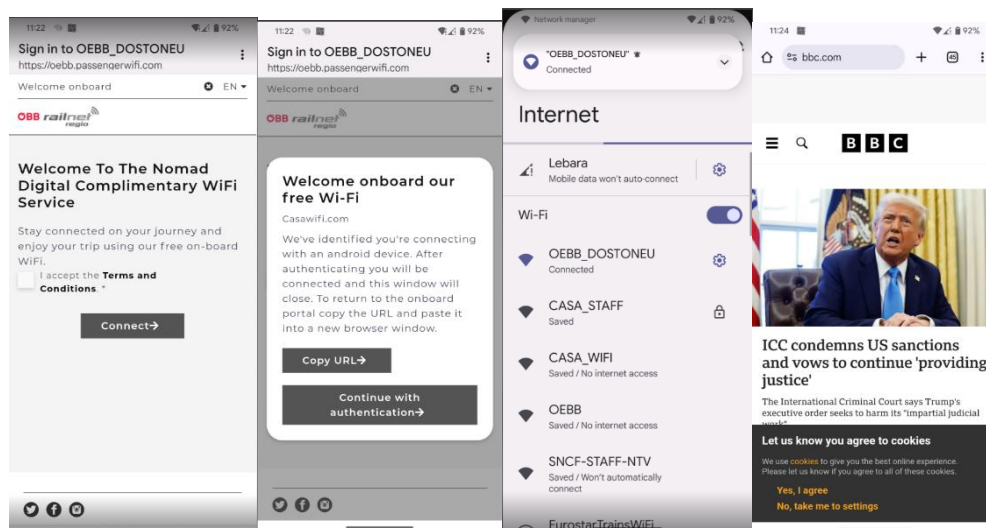
Test Device	iOS
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz
Evidence:	
	

6.3.8.2. Passenger Connectivity Flow Android Mobile

Test ID:	6.3.8.2
Description:	This test looks to follow the expected flow of an Android client device to ensure all necessary prompts are triggered and displayed when using a mobile client
Execution Steps:	The following outlines the passenger flow for connecting with an Android 1. Passenger opens Wi-Fi selection 2. Passenger selects *PROJECT_NAME* wifi 3. A prompt to sign in appears within the space of seconds 4. User clicks the prompt, and a can browser opens 5. The landing page opens and asks the user to select settings 6. The user then clicks the x in the top corner of the page 7. The user then selects the connect option that appears 8. The user then signs in 9. The user should be taken to a success page (non-existent) 10. The CNA browser will then close (the user may need to close this)

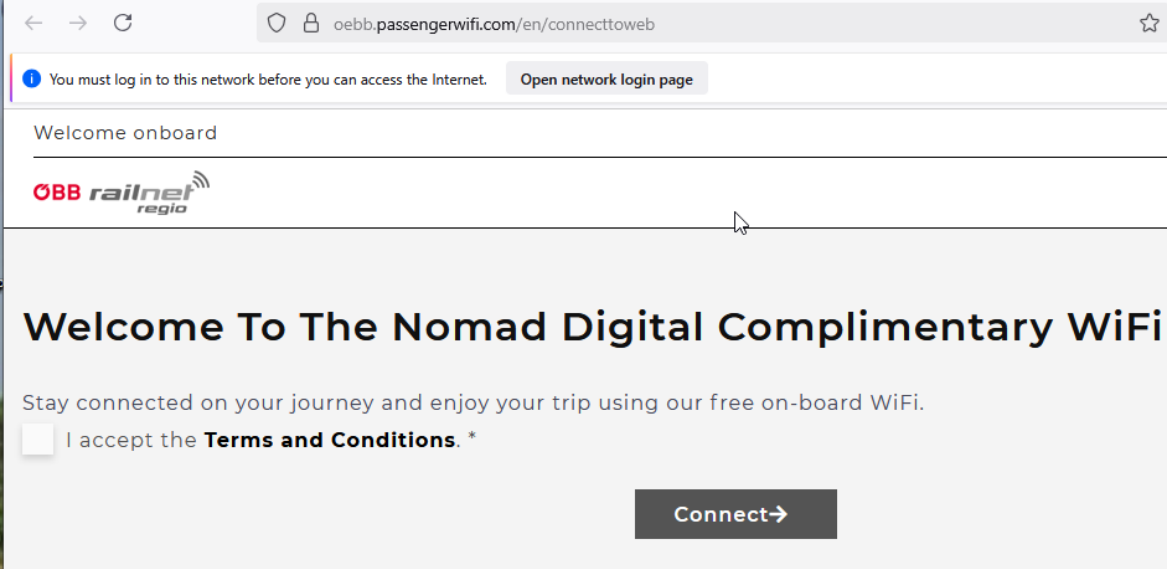
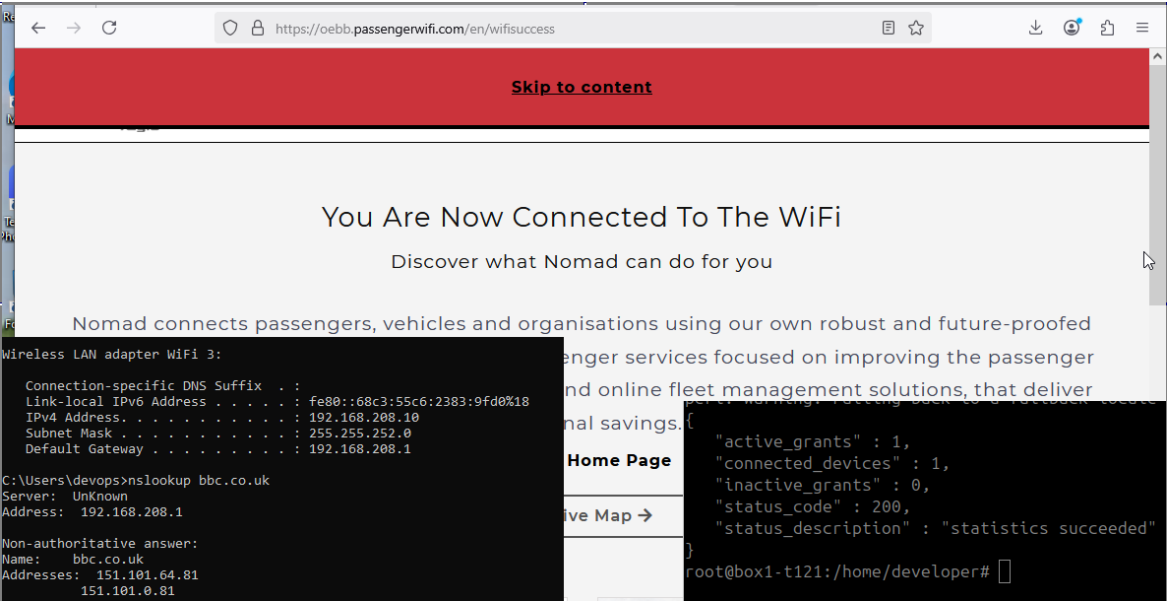
	- The user can then navigate to ifconfig.io or bbc.com or ping an external IP address - Depending on the website, the user is presented with the IP of the gateway or the bbc.co.uk website. Or users ping attempts are successful
Expected Outcome:	This test is passed when the client device can follow all steps given The client is presented with internet access on completion of the above steps
Preconditions:	System configured All APs connected are correctly configured Mar tunnel is up and is allowing connectivity
Bench:	Devops bench
Test Method:	Functional Test
Test Device	Andorid
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz

Evidence:



6.3.8.3. Passenger Connectivity Flow Using Windows Laptop/Desktop

Test ID:	6.3.8.3
Description:	This test looks to follow the expected flow of a client device to ensure all necessary prompts are triggered and displayed when using a laptop to connect to the Wi-Fi
Execution Steps:	The following outlines the passenger flow for connecting with A windows OS and Chromium Browser 1. User selects the Wi-Fi from the list of available Wi-Fi 2. A default browser should open and attempt to direct to the landing page 3. Landing page should load and prompt the User to update settings 4. The user selects the connect option from the top right of the browser 5. The page re-directs to the connecttowebsite page 6. User clicks connect and is re-directed to non-existent success page 7. The user can then navigate to ifconfig.io or bbc.com or is able to ping 8. Depending on the website, the user is presented with the IP of the gateway or the bbc.co.uk website. Or users ping attempts are successful
Expected Outcome:	This test is passed when the client device is able to follow all steps given The client is presented with internet access on completion of the above steps
Preconditions:	System configured All APs connected are correctly configured
Bench:	Devops bench
Test Method:	Functional Testing
Test Device	Windows

Execution Date	22/09/2025
Executer	Piotr Romanowski
Evidence:	
	
	

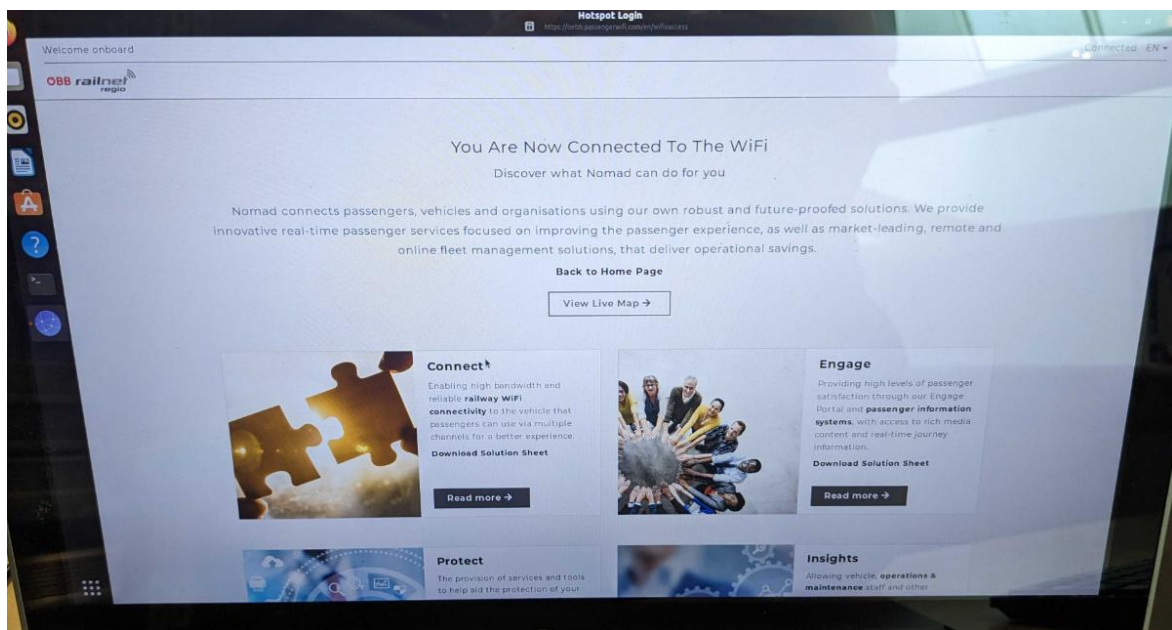
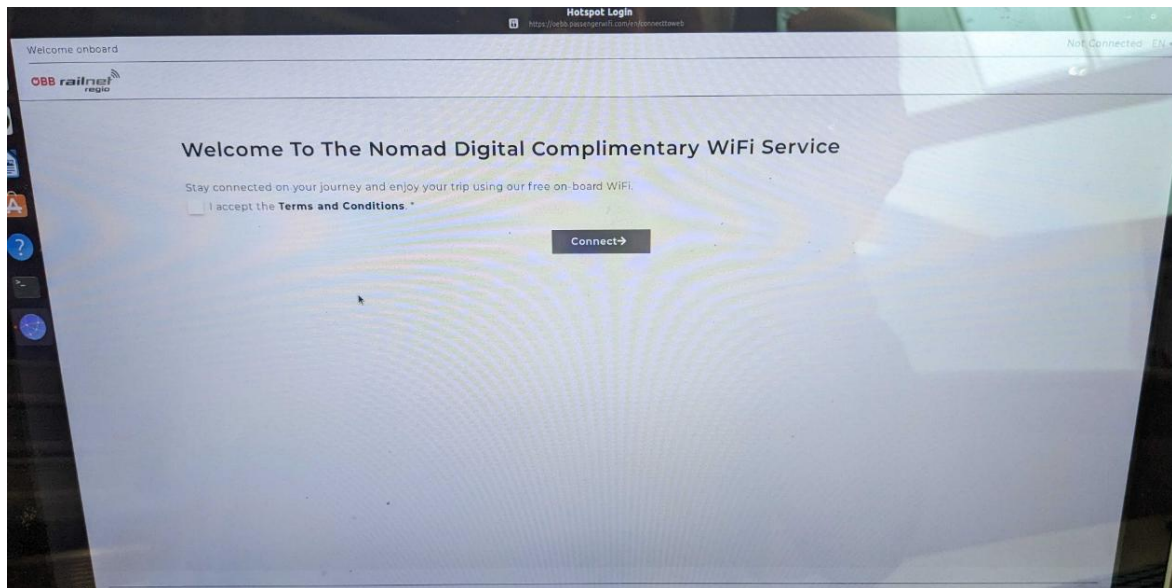
6.3.8.4. Passenger Connectivity Flow Using Linux Laptop/Desktop

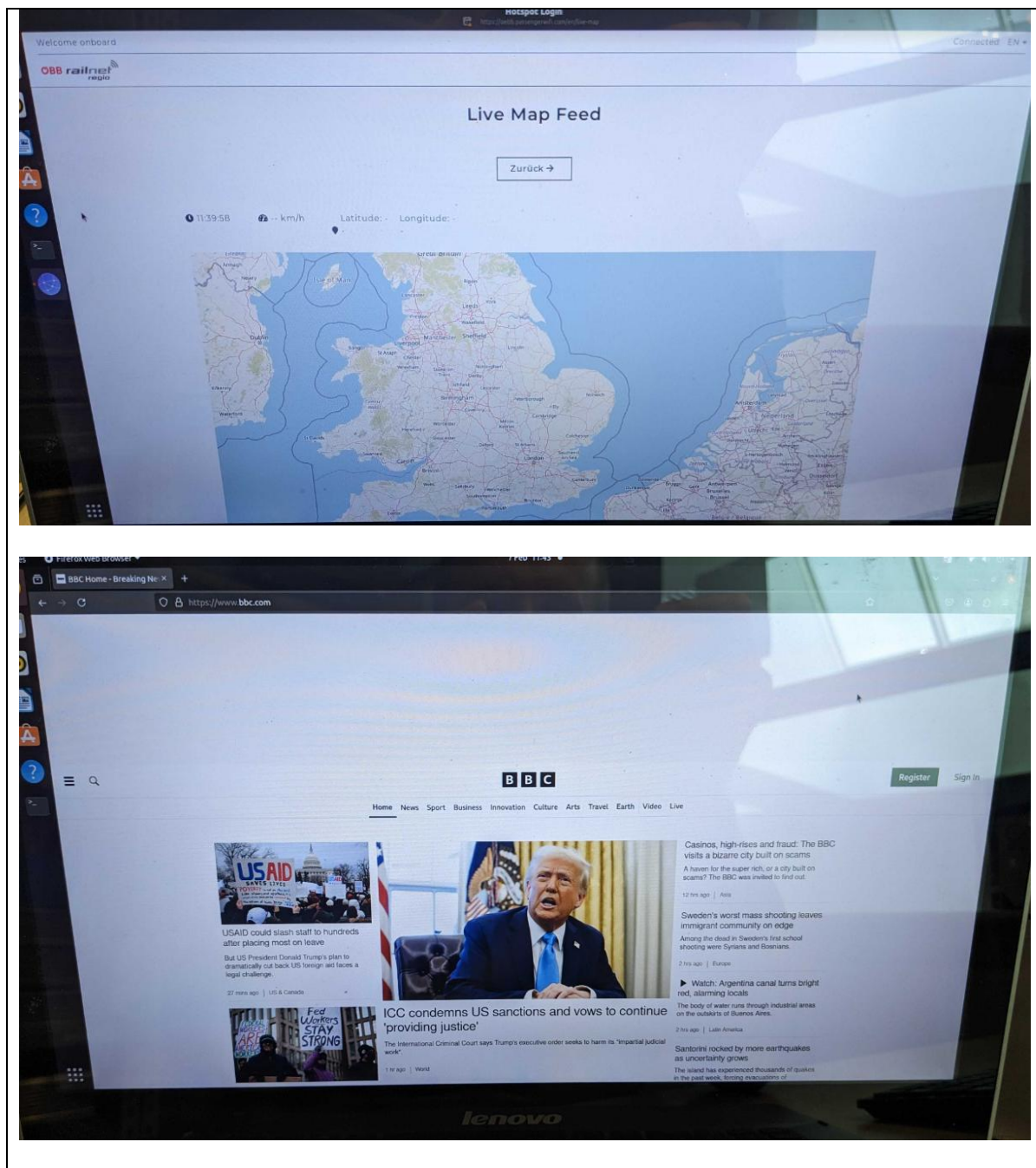
Test ID:	6.3.8.4
Description:	This test looks to follow the expected flow of a client device to ensure all necessary prompts are triggered and displayed when using a laptop to connect to the Wi-Fi
Execution Steps:	The following outlines the passenger flow for connecting with A Linux OS and Gecko web browser 1. User selects the passenger Wi-Fi network from the visible networks list. 2. A default browser should open and attempt to direct to the landing page 3. Landing page should load and prompt the User to update settings 4. The user selects the connect option from the top right of the browser 5. The page re-directs to the connecttowebsite page 6. User clicks connect and is re-directed to non-existent success page 7. The user can then navigate to ifconfig.io or bbc.com or fire ping command to 1.1.1.1/8.8.8.8 8. Depending on the website, the user is presented with the IP of the gateway or the bbc.co.uk website
Expected Outcome:	This test is passed when the client device can follow all steps given and has internet access through the CCU and its provided Wi-Fi
Preconditions:	System configured All APs connected are correctly configured
Bench:	Devops bench
Test Method:	Functional Testing
Test Device	Linux
Execution Date	07/02/2025

Executer

Szymon Dutkiewicz

Evidence:



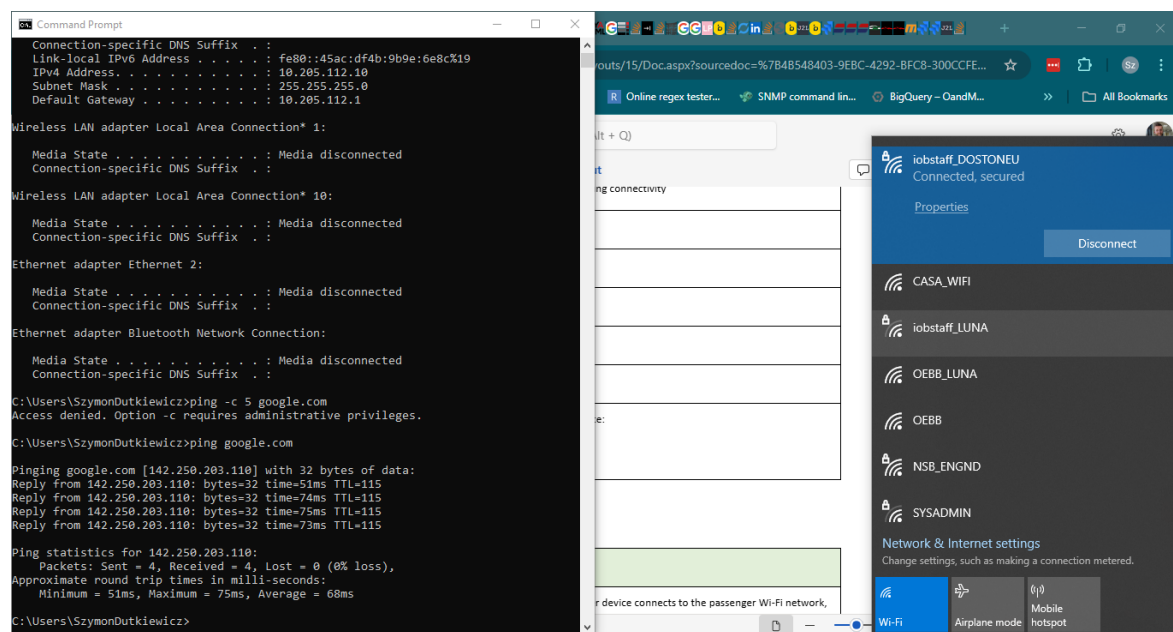


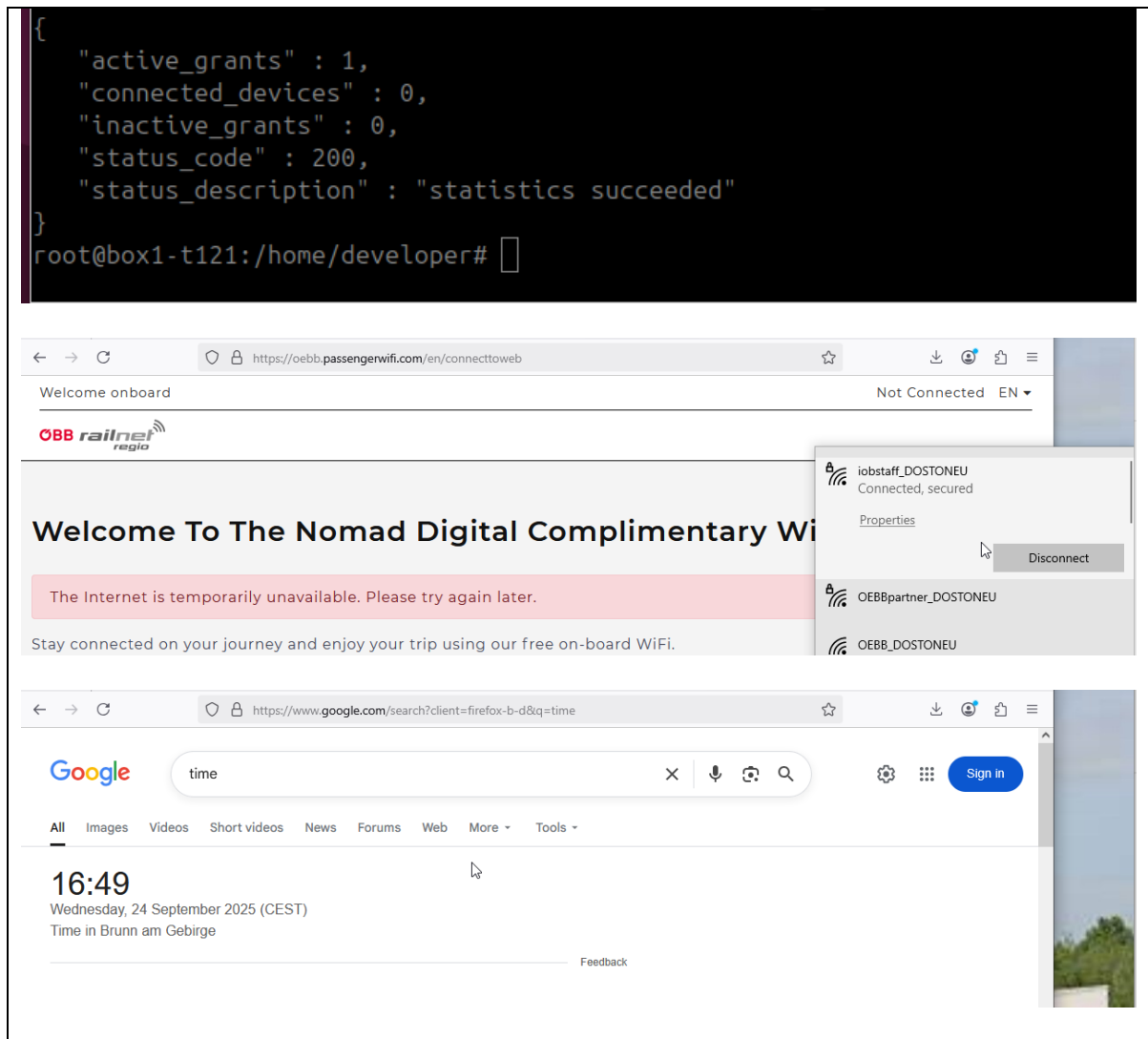
6.3.8.5. Staff Connectivity

Test ID:	6.3.8.5
Description:	Test that a client device has access to the hidden Staff SSID using the known password.
Execution Steps:	1. Open staff device Wi-Fi selection

	2. Select option to connect to hidden SSID 3. Enter the SSID 'staff' Encryption: psk2 And the known password 4. Test Device connects and is given an IP address
Expected Outcome:	This test is passed, when the test device can connect to the staff network and is issued an IP address in the correct range.
Preconditions:	CCU has been up for some time Mar tunnel is up and is allowing connectivity
Bench:	Devops bench
Test Method:	Functional testing
Test Device	Windows
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz

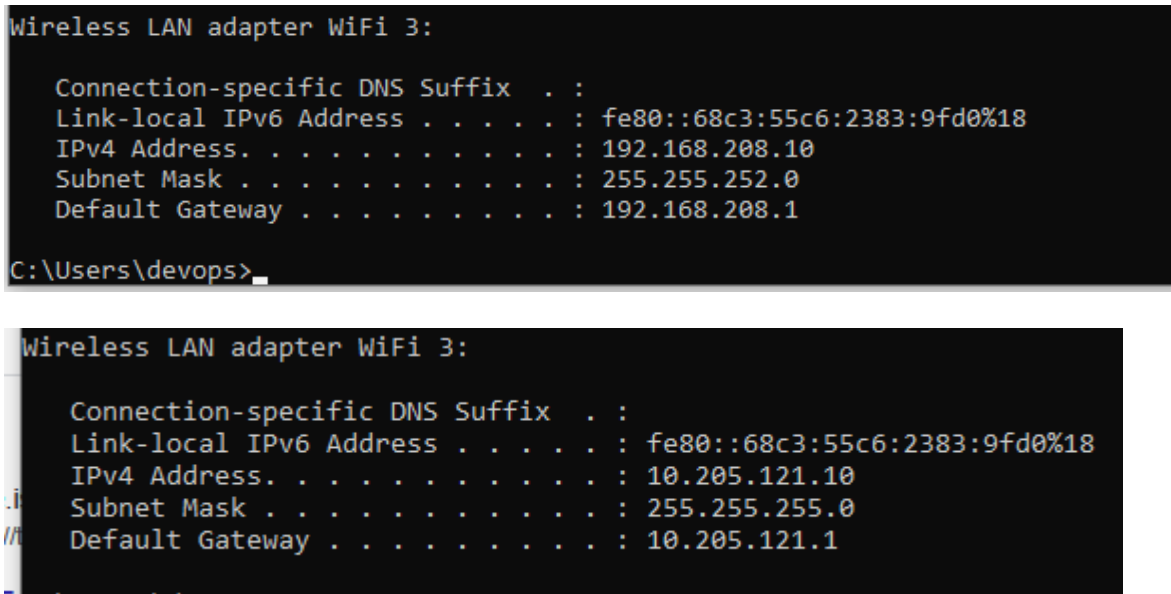
Evidence:





6.3.8.6. Passenger DHCP Functionality

Test ID:	6.3.8.6
Description:	Ensure that when a passenger device connects to the passenger Wi-Fi network, they are allocated an IP address within the expected IP range.
Execution Steps	Connect a test Device to the passenger Wi-Fi network. Once the device is connected view the IP address assigned to the device Device type: - Android: from the wi-fi list, press and hold the wi-fi network to bring up details

	- Linux: run <code>ip r</code> in terminal - Windows: Start > Settings > Network & internet > Wi-Fi - iPhone: Settings > Wi-Fi > scroll down to current Wi-Fi Check that the IP address issued to the device is in the correct range.
Expected Outcome:	Test is passed when the test device connects and gets an IP address in the correct range of 192.168.100.0/24
Preconditions:	CCU has been up for some time Mar tunnel is up and is allowing connectivity
Bench:	Devops bench
Test Method:	Functional testing
Test Device	Windows
Execution Date	19/09/2025
Executer	Piotr Romanowski
Evidence:  <pre> Wireless LAN adapter Wi-Fi 3: Connection-specific DNS Suffix . : Link-local IPv6 Address : fe80::68c3:55c6:2383:9fd0%18 IPv4 Address. : 192.168.208.10 Subnet Mask : 255.255.252.0 Default Gateway : 192.168.208.1 C:\Users\devops> Wireless LAN adapter Wi-Fi 3: Connection-specific DNS Suffix . : Link-local IPv6 Address : fe80::68c3:55c6:2383:9fd0%18 IPv4 Address. : 10.205.121.10 Subnet Mask : 255.255.255.0 Default Gateway : 10.205.121.1 </pre>	

6.3.8.7. Staff DHCP Functionality

Test ID:	6.3.8.7
Description:	Ensure that when a staff device connects to the staff Wi-Fi network, they are allocated an IP address within the expected IP range.
Execution Steps:	Connect test Device to the Staff Wi-Fi and enter the password if not retained from test 6.3.8.5. Once the device is connected view the IP address assigned to the device Device type: - Android: from the wi-fi list, press and hold the wi-fi network to bring up details - Linux: run ip r in terminal - Windows: Start > Settings > Network & internet > Wi-Fi - iPhone: Settings > Wi-Fi > scroll down to current Wi-Fi Check that the IP address issued to the device is in the correct range.
Expected Outcome:	Test is passed when the test device connects and gets an IP address in the correct range
Preconditions:	CCU has been up for some time Mar tunnel is up and is allowing connectivity
Bench:	Devops bench
Test Method:	Functional testing
Test Device	Windows
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz

Evidence:

```
Command Prompt

C:\Users\SzymonDutkiewicz>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::45ac:df4b:9b9e:6e8c%19
    IPv4 Address. . . . . : 10.205.112.10
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 10.205.112.1

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

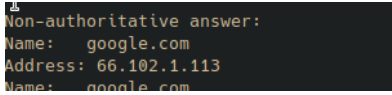
Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

C:\Users\SzymonDutkiewicz>
```

6.3.8.8. DNS Functionality

Test ID:	6.3.8.8
Description:	Validate that DNS queries are resolved from the CCU. Queries can be resolved for a long period of time.
Execution Steps:	Access the CCU either via SSH or console

	Confirm that the CCU is connected to backend by running 'marcli' <i>Check for ACTIVE state on modem</i> <i>Can also be confirmed with ping 8.8.8.8 with no packet loss</i> Run command 'nslookup google.com' The command should return a series of IP addresses in a similar format to that given below  Repeat test with: bbc.com, bbc.co.uk, itv.com Test may also be completed by pinging domains and those pings respond with IPs
Expected Outcome:	CCU can resolve website addresses consistently for multiple websites The CCU reports back a single IP or multiple IPs for the domains given
Preconditions:	CCU is online and staged CCU has active connection to backend Different domains can be used
Bench:	Devops bench
Test Method:	Functional testing
Test Device	Linux
Execution Date	22/09/2025
Executer	Szymon Dutkiewicz / Piotr Romanowski
Evidence:	

```
root@box1-t112:~# nslookup google.com
Server:      127.0.0.1
Address:     127.0.0.1#53
```

```
Non-authoritative answer:
Name:   google.com
Address: 172.217.168.78
Name:   google.com
Address: 2a00:1450:4001:831::200e
```

```
root@box1-t112:~# nslookup bbc.com
Server:      127.0.0.1
Address:     127.0.0.1#53
```

```
Non-authoritative answer:
Name:      bbc.com
Address:   151.101.128.81
Name:      bbc.com
Address:   151.101.0.81
Name:      bbc.com
Address:   151.101.64.81
Name:      bbc.com
Address:   151.101.192.81
Name:      bbc.com
Address:   2a04:4e42:400::81
Name:      bbc.com
Address:   2a04:4e42::81
Name:      bbc.com
Address:   2a04:4e42:200::81
Name:      bbc.com
Address:   2a04:4e42:600::81
```

```
root@box1-t112:~#
```

```

root@kali:~# cat /etc/hosts
127.0.0.1 localhost
::1 localhost
root@kali:~# root@box1-t121:/home/developer# for i in bbc.co.uk itv.com google.com; do nslookup $i | grep -Ei "(address|name)"; done
Address:      127.0.0.1#53
Name:   bbc.co.uk
Address: 151.101.64.81
Name:   bbc.co.uk
Address: 151.101.192.81
Name:   bbc.co.uk
Address: 151.101.0.81
Name:   bbc.co.uk
Address: 151.101.128.81
Address:      127.0.0.1#53
Name:   itv.com
Address: 75.2.43.168
Name:   itv.com
Address: 99.83.221.243
Address:      127.0.0.1#53
Name:   google.com
Address: 142.250.74.206

```

6.3.8.9. DNS response time

Test ID:	6.3.8.9
Description:	Ensure that DNS queries are resolved in a timely manner, a high response time on DNS is indicative of upstream DNS issue
Execution Steps:	Access the CCU either via SSH or console Confirm that the CCU is connected to backend by running 'marcli' <i>Check for ACTIVE state on modem</i> <i>Can also be confirmed with ping 8.8.8.8 with no packet loss</i> Run command 'time host diffchecker.com' The command should return a series of IP addresses and the time taken to retrieve the DNS entry Repeat test with: bbc.com, bbc.co.uk, itv.com Ensure that the DNS responds in 0.20s for all entries
Expected Outcome:	CCU can resolve website addresses consistently for multiple websites in a timely manner
Preconditions:	CCU is online and staged CCU has active connection to backend Different domains can be used
Bench:	Devops bench
Test Method:	Functional testing
Test Device	Linux
Execution Date	22/09/2025
Executer	Szymon Dutkiewicz / Piotr Romanowski

Evidence:

```
root@box1-t112:~# time host diffchecker.com
diffchecker.com has address 188.114.97.3
diffchecker.com has address 188.114.96.3
diffchecker.com has IPv6 address 2a06:98c1:3120::3
diffchecker.com has IPv6 address 2a06:98c1:3121::3
diffchecker.com mail is handled by 30 aspmx5.googlemail.com.
diffchecker.com mail is handled by 30 aspmx2.googlemail.com.
diffchecker.com mail is handled by 10 aspmx.l.google.com.
diffchecker.com mail is handled by 20 alt1.aspmx.l.google.com.
diffchecker.com mail is handled by 20 alt2.aspmx.l.google.com.
diffchecker.com mail is handled by 30 aspmx4.googlemail.com.
diffchecker.com mail is handled by 30 aspmx3.googlemail.com.

real    0m0.195s
user    0m0.006s
sys     0m0.006s
root@box1-t112:~# time host bbc.com
bbc.com has address 151.101.64.81
bbc.com has address 151.101.128.81
bbc.com has address 151.101.192.81
bbc.com has address 151.101.0.81
bbc.com has IPv6 address 2a04:4e42:600::81
bbc.com has IPv6 address 2a04:4e42:400::81
bbc.com has IPv6 address 2a04:4e42::81
bbc.com has IPv6 address 2a04:4e42:200::81
bbc.com mail is handled by 20 cluster8a.eu.messagelabs.com.
bbc.com mail is handled by 10 cluster8.eu.messagelabs.com.

real    0m0.354s
user    0m0.006s
sys     0m0.006s
root@box1-t112:~#
```

```

root@box1-t121:/home/developer# time host diffchecker.com
diffchecker.com has address 104.26.5.140
diffchecker.com has address 104.26.4.140
diffchecker.com has address 172.67.71.147
diffchecker.com mail is handled by 30 aspmx2.googlemail.com.
diffchecker.com mail is handled by 10 aspmx.l.google.com.
diffchecker.com mail is handled by 20 alt1.aspmx.l.google.com.
diffchecker.com mail is handled by 30 aspmx4.googlemail.com.
diffchecker.com mail is handled by 30 aspmx5.googlemail.com.
diffchecker.com mail is handled by 20 alt2.aspmx.l.google.com.
diffchecker.com mail is handled by 30 aspmx3.googlemail.com.
^C
real    0m0.210s
user    0m0.012s
sys     0m0.000s
root@box1-t121:/home/developer# time host bbc.com
bbc.com has address 151.101.128.81
bbc.com has address 151.101.0.81
bbc.com has address 151.101.64.81
bbc.com has address 151.101.192.81
bbc.com mail is handled by 10 cluster8.eu.messagelabs.com.
bbc.com mail is handled by 20 cluster8a.eu.messagelabs.com.
^C
real    0m0.186s
user    0m0.015s
sys     0m0.000s
root@box1-t121:/home/developer#

```

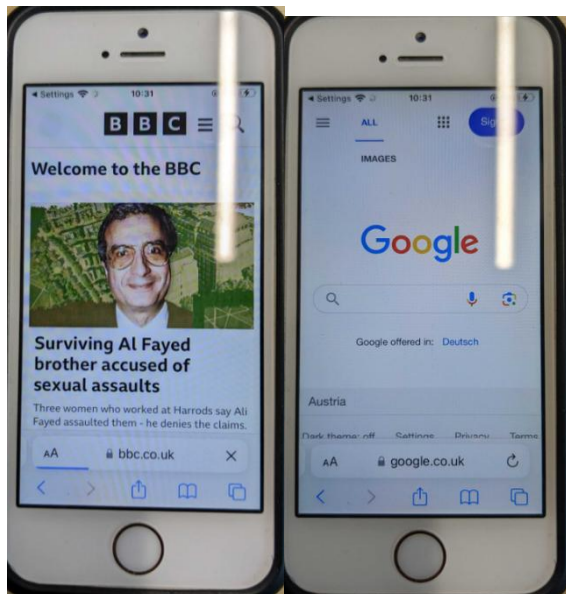
6.3.8.10. Passenger Network DNS Resolution

Test ID:	6.3.8.10
Description:	Validate that DNS queries are resolved from connected client devices. For this test we connect a client to the passenger network and direct it to various http sites over a period of five minutes.
Execution Steps	1. Tester connects a test device to the Wi-fi through the passenger SSID 2. User flows sign-in steps outlined in 6.3.8.{1-2} <i>The user has signed into the Wi-Fi through the portal</i> 3. User attempts to navigate to a webpage such as bbc.com, google.com example.com 4. The test device can resolve these URLs and webpages can load Test device may also use DNS lookup tools such as nslookup (windows, linux, mac) Test device may also use Network test tool Applications to lookup DNS

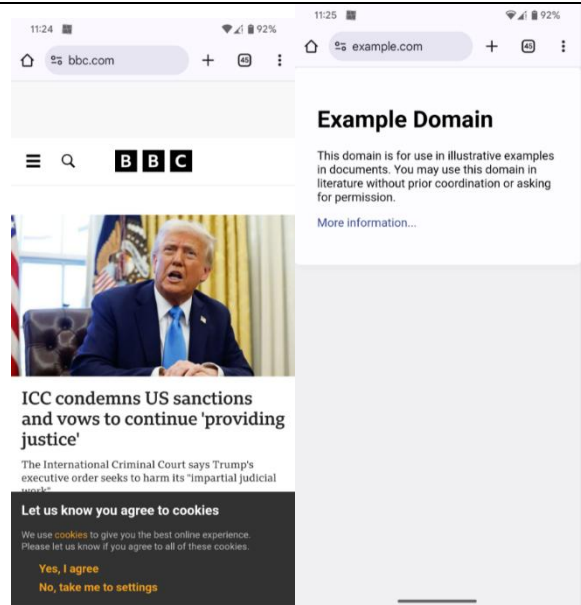
Expected Outcome:	Test is considered passed when the Test device can resolve the DNS of websites by either looking up their correlating IP or viewing the webpage
Preconditions:	CCU is online and is connected to backend Test Steps in 6.3.8.{1-2} have been passed
Bench:	Devops bench
Test Method:	Functional testing
Test Device	IOS, Android, Windows, Linux
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz

Evidence:

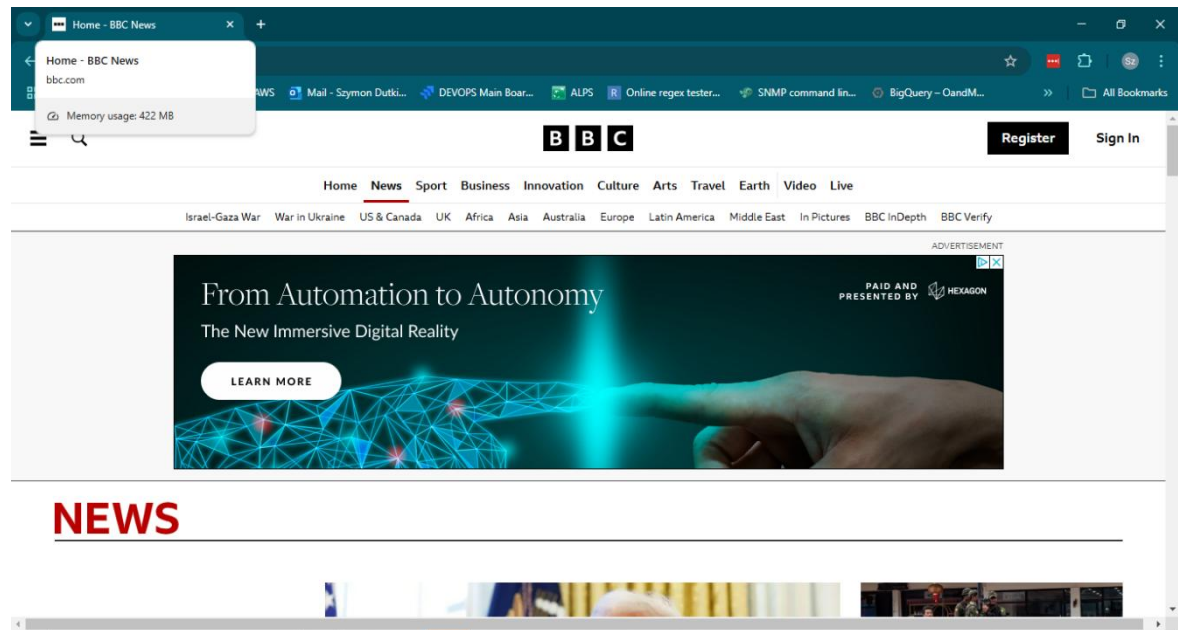
IOS

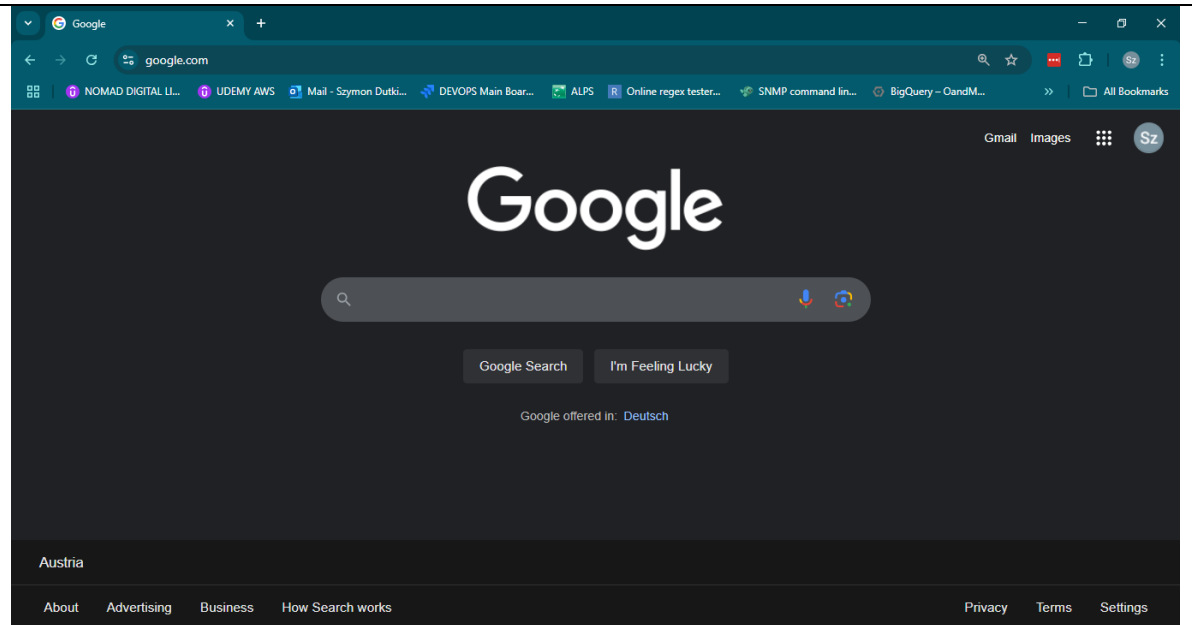


Android

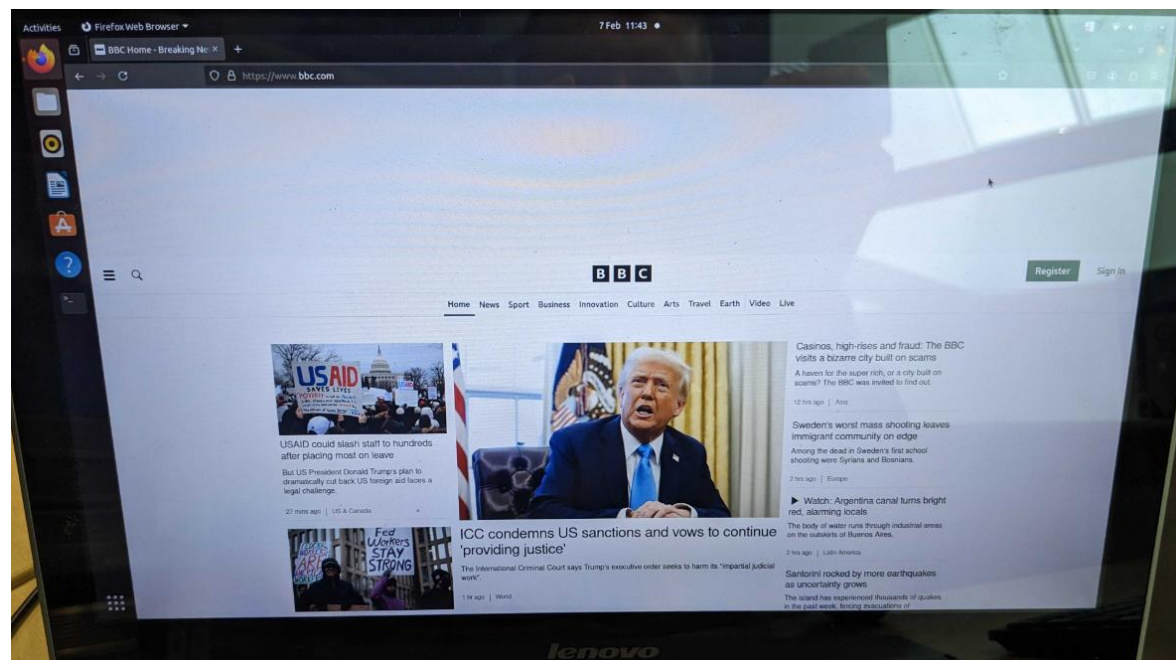


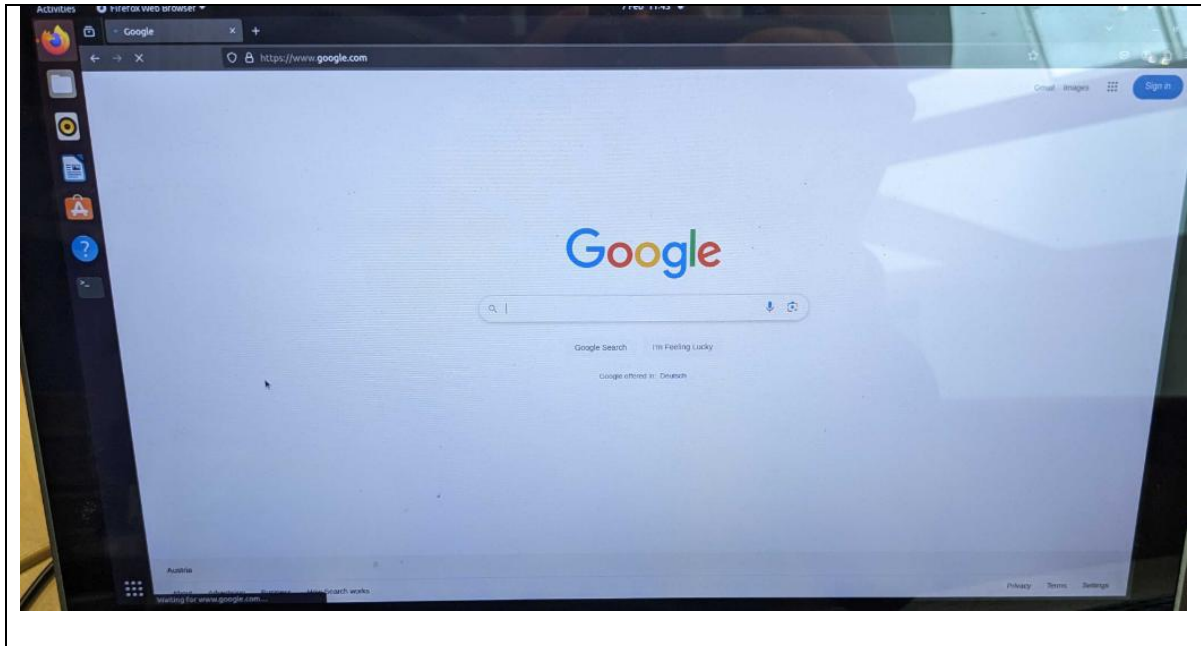
Windows





Linux



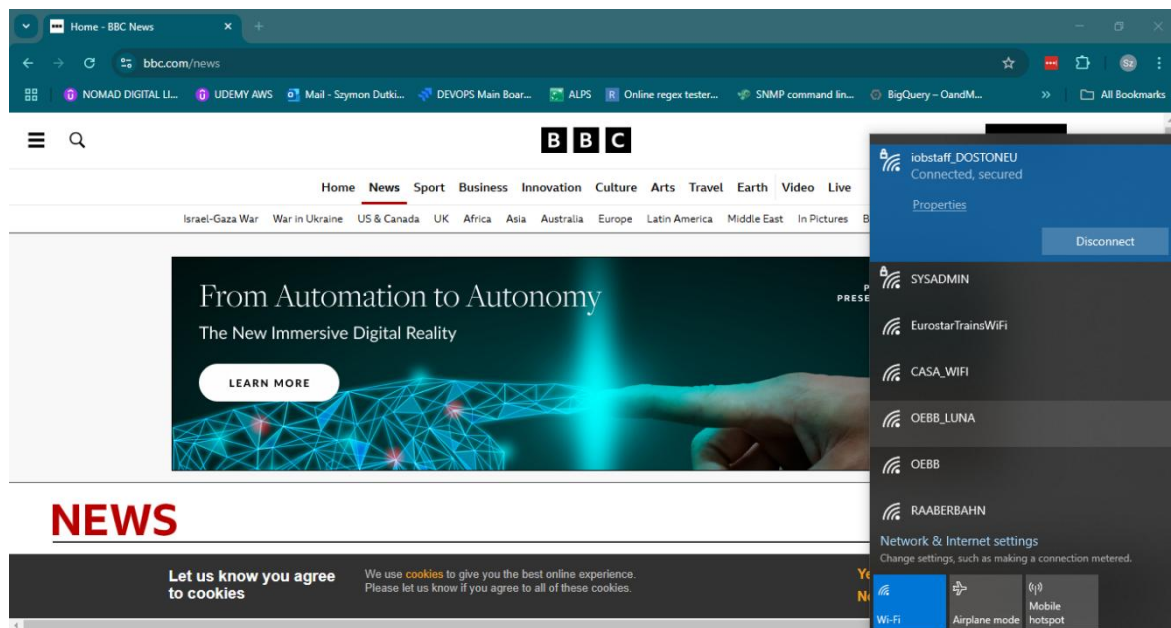


6.3.8.11. Staff Network DNS Resolution

Test ID:	6.3.8.11
Description:	Validate that DNS queries are resolved from connected staff devices. For this test we connect a client to the staff network and direct it to various http sites over a period of five minutes.
Execution Steps:	1. Tester connects a test device to the Wi-fi through the staff SSID <i>Enter SSID password</i> 2. User attempts to navigate to a webpage such as bbc.com, google.com example.com 3. The test device can resolve these URLs and webpages can load Test device may also use DNS lookup tools such as nslookup (windows, linux, mac) Test device may also use Network test tool Applications to lookup DNS

Expected Outcome:	Test device can resolve URLs and or load webpages via URL through the staff network.
Preconditions:	CCU is online and is connected to backend
Bench:	Devops bench
Test Method:	Functional testing
Test Device	Windows
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz

Evidence:

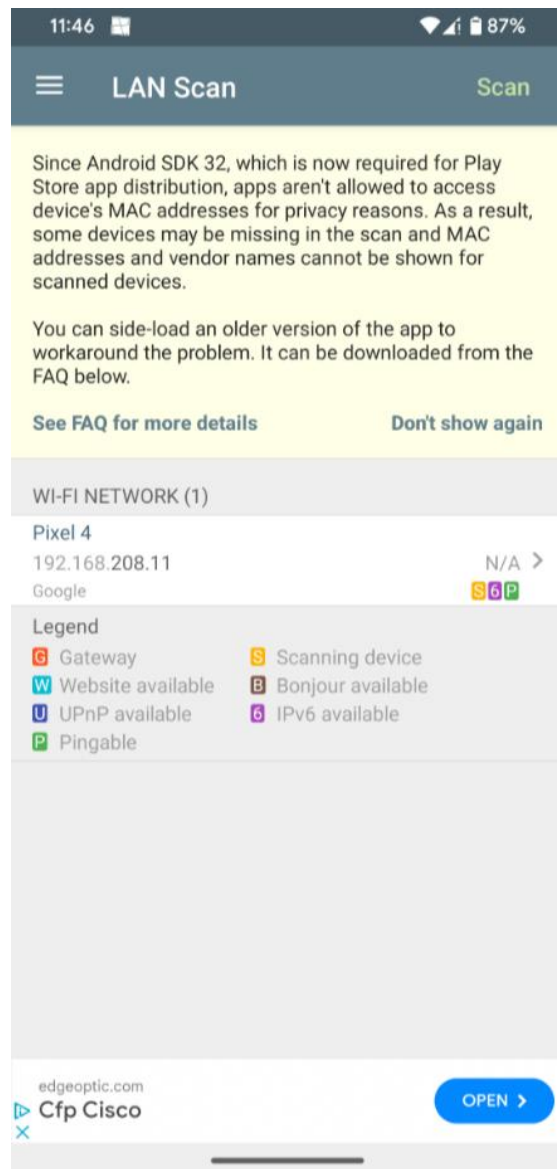


6.3.8.12. Client Isolation Functionality

Test ID:	6.3.8.13
Description:	This test looks to verify that passengers are unable to see each other when performing packet snooping or network scans.

Execution Steps:	Connect two devices to the passenger Wi-Fi network. This can be the same AP or different AP's. Begin running a network scan using the Net Analyzer app or arp DO NOT run both scans simultaneously (leaves echos) Check that the only devices visible are the gateways and the device performing the scanning
Expected Outcome:	This test is passed, when two or more client test devices are connected to the access points/CCU and that a test device running a network scan is only able to see the CCU or Portal and NOT other test clients
Preconditions:	System configured All APs connected are correctly configured Two or more test client devices are connected to APs
Bench:	Devops bench
Test Method:	Functional Testing
Test Device	Linux, Android
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz
Evidence:	

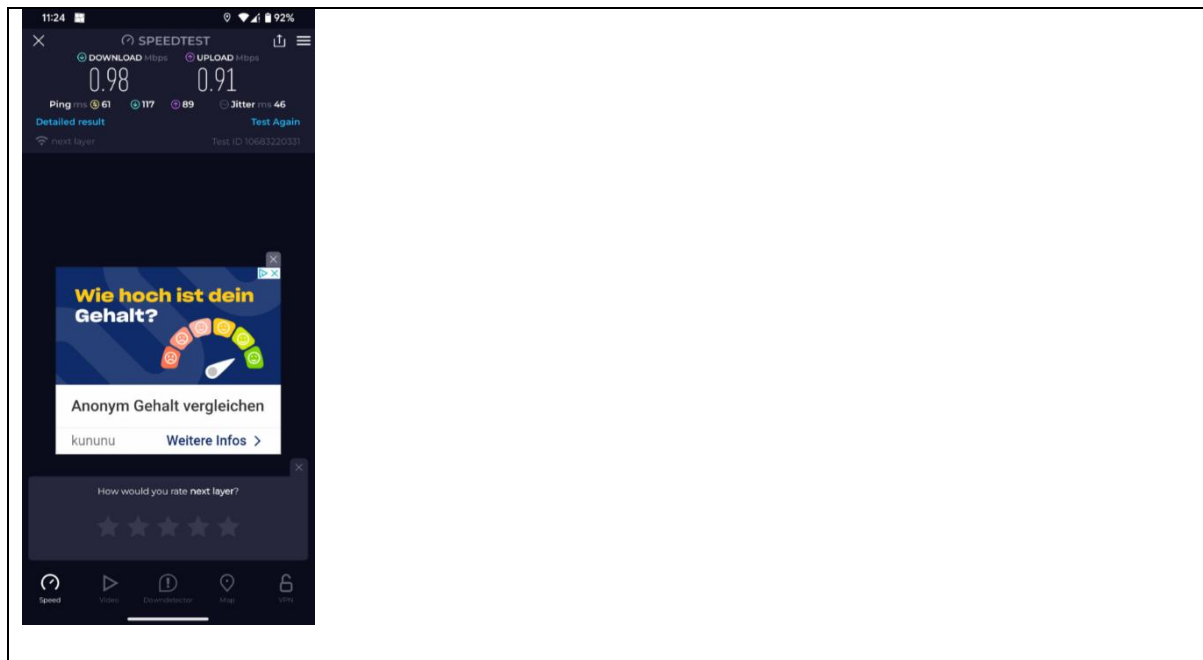

```
devops@devops-Lenovo-Flex-2-14:~$ ip r
default via 192.168.208.1 dev wlp2s0 proto dhcp metric 600
169.254.0.0/16 dev wlp2s0 scope link metric 1000
192.168.208.0/22 dev wlp2s0 proto kernel scope link src 192.168.208.12 metric 600
devops@devops-Lenovo-Flex-2-14:~$ nmap -sP 192.168.208.1/22
Starting Nmap 7.80 ( https://nmap.org ) at 2025-02-07 11:44 GMT
Nmap scan report for _gateway (192.168.208.1)
Host is up (0.050s latency).
Nmap scan report for devops-Lenovo-Flex-2-14 (192.168.208.12)
Host is up (0.00011s latency).
Nmap done: 1024 IP addresses (2 hosts up) scanned in 17.71 seconds
devops@devops-Lenovo-Flex-2-14:~$
```



6.3.8.13. Performance – Passenger Rate Limiting & Throttling

Test ID:	6.3.8.13
Description:	The purpose of this test is to validate that the user is rate limited / throttled after exceeding an established download threshold
Execution Steps:	1. Log into NOMAD CONNECT as {developer} 2. Run<pre>{watch marcli all}</pre>and confirm that all Modems are active and that NOMAD CONNECT tunnel is connected (the tunnel is connected if at least one of the links is in ACTIVE state) 3. Ensure the mar3-captiveportal-api settings are configured correctly. The /etc/mar3-captiveportal-api/service.json file determines what the user limits and fair use quotas are. For this test configure it as follows:<pre>[{ "id": 1, "name": "standard", "reset_interval": 0, "initial_prio_class": 1, "initial_bw_down": 100000000, "initial_bw_up": 100000000, "limits": [{ "fair_use": 1048576, "when_reached_bw_down": 50000, "when_reached_bw_up": 50000, "when_reached_prio_class": 1 }] }]</pre>Above configuration file will ensure that a user initially has 100 Mbit /s up/down of throughput available, once the user hits 10 MB of fair use quota they will be throttled to 50 kbit/s up/down. 4. After the configuration file has been modified, restart the captive portal api with the systemctl restart mar3-captiveportal-api command. Client device throttling: 1. Connect a client device to MAR AP SSID and validate internet connectivity 2. Trigger the following download from client device:

	http://noc-demo.21net.com/1GB.bin 3. Validate that client device throughput is rate limited when download threshold is exceeded, this can be checked with the following commands: <i>cp-list.sh: show a list of all sessions</i> <i>cp-status.sh <ip>: show the status of a specific session</i> Negative test - Validate other users are not affected: 1. Connect a second client device & validate that the new user download rate cap has NOT been applied to new device.
Expected Outcome:	Upon reaching the download limit, user is throttled back to the predefined threshold
Preconditions:	System configured Switch and AP configured Enough throughput on the CCU to be able to reach defined values
Bench:	Devops bench
Test Method:	Functional Testing
Test Device	Android
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz
Evidence:	

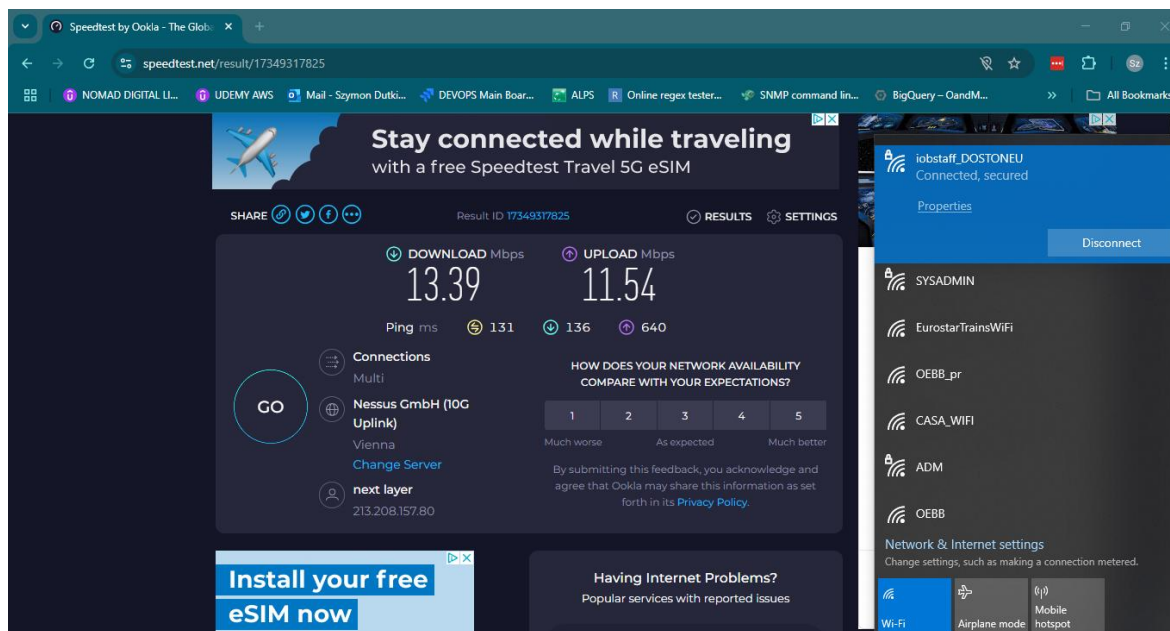


6.3.8.14. Performance – Staff Wi-Fi speed test

Test ID:	6.3.8.14
Description:	Test performance of Staff wi-fi network to establish AP performance and general performance
Execution Steps:	Connect test device to the staff Wi-fi Enter password provided Ensure you have internet connectivity by going to google.com Perform speed test - Google's search speedtest - speedtest.net - A speedtest App on the test device Ensure test result report at least 1 mbps download and upload
Expected Outcome:	This test is passed when the client device can achieve a download rate of 1mbps+ is achieved and an upload rate of 1mbps
Preconditions:	System configured and sync'd with puppet

	Mar Tunnel is active with the default device or at least two modems with SIMs Power cycled CCU
Bench:	Devops bench
Test Method:	Functional testing
Test Device	Windows
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz

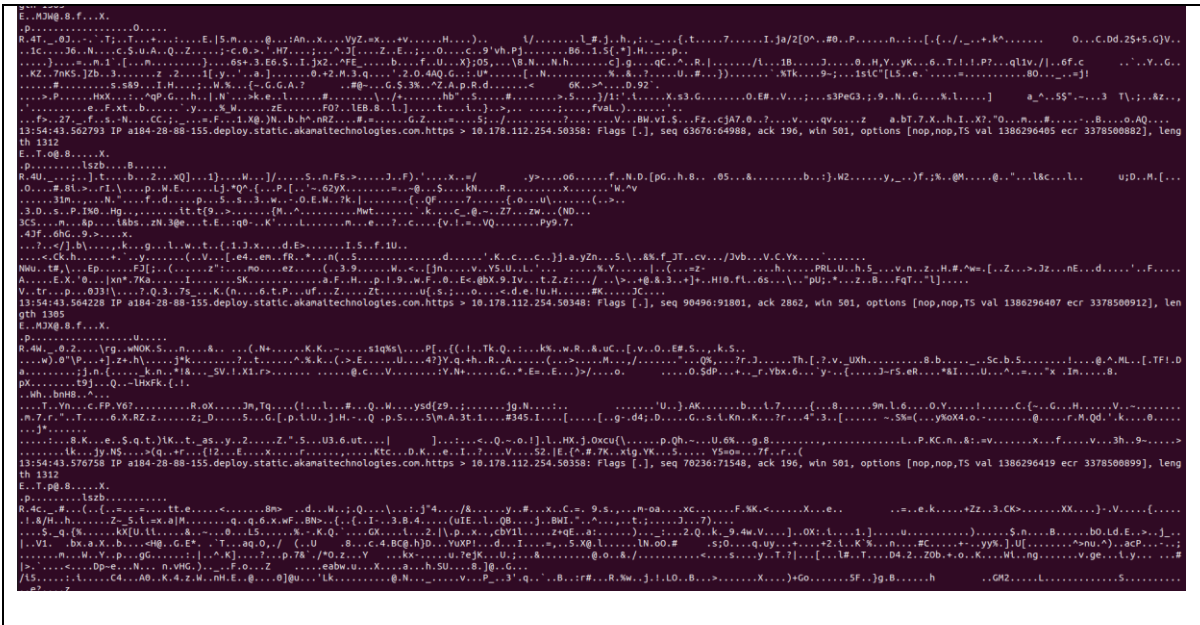
Evidence:



6.3.8.15. Passenger Traffic Encryption

Test ID:	6.3.8.15
Description:	Test that all passenger traffic is encrypted
Execution Steps:	Connect client device to Wi-fi (either AP) Begin browsing or streaming websites

	In a Linux Terminal, connect to the backend the unit is pointed at 185.201.63.19 Begin running a tcpdump for the connection from the CCU <i>tcpdump -As0 -i any '(dst 10.167.113.254 or src 10.167.113.254)' and not port '(22 or 8883 or 10050 or 10051)'</i> Confirm that data is not readable
Expected Outcome:	This test is passed, when the traffic originating from the CCU to CDC is encrypted
Preconditions:	All previous tests passed, all APs and switches connected to bench Client tests passed Access to the backend is possible
Bench:	Devops
Test Method:	Functional Testing
Test Device	Linux
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz
Evidence:	



6.3.8.16. Passenger Network Firewalling

Test ID:	6.3.8.16
Description:	Verify if appropriate security measures were taken to secure the passenger network from gaining access to the CCU, by blocking access to all unnecessary ports and other networks
Execution Steps:	Connect client device to Passenger Wi-fi 1. Run a scan on all ports to verify which ports are open Linux: <i>nmap -p- *CCU GW IP*</i> Android: using apps such as FPing or Net Analyzer 2. Ping CCU global IP, ping CCU GW for Staff network
Expected Outcome:	Only 53, 80 & 443 ports should be visible to the device. Unable to ping any other networks on the CCU.
Preconditions:	Security package deployed to bench

Bench:	Devops
Test Method:	Functional Testing
Test Device	Windows
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz

Evidence:

```
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::45ac:df4b:9b9e:6e8c%19
    IPv4 Address. . . . . : 192.168.208.13
    Subnet Mask . . . . . : 255.255.252.0
    Default Gateway . . . . . : 192.168.208.1

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\SzymonDutkiewicz>nmap -p- 192.168.208.1
Starting Nmap 7.80 ( https://nmap.org ) at 2025-02-07 13:42 GMT Standard Time
Nmap scan report for 192.168.208.1
Host is up (0.068s latency).
Not shown: 65532 filtered ports
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 00:21:21:21:01:01 (VRmagic GmbH)

Nmap done: 1 IP address (1 host up) scanned in 488.75 seconds

C:\Users\SzymonDutkiewicz>
```


6.3.8.17. Staff Network Firewalling

Test ID:	6.3.8.17
Description:	Verify if appropriate security measures were taken to secure the staff network from gaining access to the CCU, by blocking access to all unnecessary ports and other networks
Execution Steps:	Connect client device to Staff Wi-fi 1. Run a scan on all ports to verify which ports are open Linux: <i>nmap -p- *CCU GW IP*</i> Android: using apps such as Fping or Net Analyzer 2. Ping CCU global IP, ping CCU GW for Passenger network
Expected Outcome:	Only 53, 80 & 443 ports should be visible to the device. Unable to ping any other networks on the CCU.
Preconditions:	Security package deployed to bench
Bench:	Devops
Test Method:	Functional Testing
Test Device	Windows
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz
Evidence:	

```
Connection-specific DNS Suffix . : 
Link-local IPv6 Address . . . . . : fe80::45ac:df4b:9b9e:6e8c%19
IPv4 Address. . . . . : 10.205.112.10
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 10.205.112.1

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection* 10:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

C:\Users\SzymonDutkiewicz>nmap -p- 10.205.112.1
Starting Nmap 7.80 ( https://nmap.org ) at 2025-02-07 13:11 GMT Standard Time
Nmap scan report for 10.205.112.1
Host is up (0.055s latency).
Not shown: 65530 filtered ports
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp    open  https
2947/tcp   closed gpsd
8005/tcp   open  mxi
MAC Address: 00:21:21:21:03:01 (VRmagic GmbH)

Nmap done: 1 IP address (1 host up) scanned in 448.69 seconds

C:\Users\SzymonDutkiewicz>
```

6.3.9. Portal functionality

6.3.9.1. Docker service autostart

Test ID:	6.3.9.1
Description:	The portal is hosted on the CCU using Docker, docker must start on CCU startup. The registry image of docker must start itself
Execution Steps:	5. Power cycle the CCU to ensure a fresh boot 6. Log into the CCU using either console or SSH access see previous testing 7. Once on CCU escalate privileges using 'sudo su' 1. Run command 'systemctl status docker' 2. Process should report back active and running 3. Running docker ps shows a registry running May take up to two minutes after <i>power cycle</i>
Expected Outcome:	The docker service and registry start as part of CCU boot sequence
Preconditions:	Base NC image is sync'd with puppet Access through SSH or serial to CCU
Bench:	Devops bench
Test Method:	Functional testing
Test Device	Linux
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz
Evidence:	

```
root@box1-t112:/home/developer# systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/lib/systemd/system/docker.service; enabled; preset: disabled)
   Active: active (running) since Fri 2025-02-07 14:34:11 UTC; 39min ago
   TriggeredBy: @docker.socket
     Docs: https://docs.docker.com
    Main PID: 2727 (dockerd)
      Tasks: 77 (limit: 76835)
     Memory: 1.3G
        CPU: 41.78ss
    CGroup: /system.slice/docker.service
            └─2727 /usr/bin/dockerd --data-root=/var/tmp/docker_data --exec-root=/var/tmp/docker_exec --ip-masq=false --default-ulimit core=1024 --default-ulimit nice=0 --default-ulimit nofile=1024:2048
            └─1216 containerd --config /var/tmp/docker_exec/containerd/containerd.toml --log-level info
            └─1825 /usr/bin/docker-proxy -proto tcp -host-ip 127.0.0.1 -host-port 5000 -container-ip 172.15.0.2 -container-port 5000
            └─1864 /usr/bin/containerd-shim-runc.v2 --namespace moby --id 1604ddb87dc5bd2474a6e6cfd8c3bc5a9b1d4a39fe298ed50798ca7b666d339 --address /var/tmp/docker_exec/containerd/containerd.sock
            └─1852 /usr/bin/docker-proxy -proto tcp -host-ip 0.0.0.0 -host-port 443 -container-ip 172.15.0.3 -container-port 443
            └─1899 /usr/bin/containerd-shim-runc.v2 --namespace moby --id 15382648fc131912c407b54dc535b64ef9a1cb0b5bfab077f6d7c8daf11d140 --address /var/tmp/docker_exec/containerd/containerd.sock

Feb 07 14:34:14 box1-t112 dockerd[3216]: ttime="2025-02-07T14:34:14.654591450Z" level=info msg="loading plugin \"io.containerd.trrpc.v1.task\"..." runtime=io.containerd.runc.v2 type=io.containerd.trrpc.v1
Feb 07 14:34:14 box1-t112 dockerd[3216]: ttime="2025-02-07T14:34:14.654740660Z" level=info msg="starting signal loop" namespace=moby path=/var/tmp/docker_exec/containerd/daemon/io.containerd.runtime.v2.1
Feb 07 14:34:15 box1-t112 dockerd[2727]: ttime="2025-02-07T14:34:15.581928045Z" level=warning msg="Error getting v2 registry: Get \"https://localhost:5000/v2/\": http: server gave HTTP response to HTTPS re
Feb 07 14:34:29 box1-t112 dockerd[2727]: ttime="2025-02-07T14:35:29.004668520Z" level=info msg="Attempting next endpoint for pull after error: Get \"https://localhost:5000/v2/\": http: server gave HTTP re
Feb 07 14:35:29 box1-t112 dockerd[2727]: ttime="2025-02-07T14:35:29.004668520Z" level=info msg="No non-localhost DNS nameservers are left in resolv.conf. Using default external servers: [nameserver 8.8.8.8
Feb 07 14:35:29 box1-t112 dockerd[2727]: ttime="2025-02-07T14:35:29.004699699Z" level=info msg="IPv6 enabled: Adding default IPv6 external servers: [nameserver 2001:4860:4860::8888 nameserver 2001:4860:4
Feb 07 14:35:29 box1-t112 dockerd[3216]: ttime="2025-02-07T14:35:29.030063147Z" level=info msg="loading plugin \"io.containerd.event.v1.publisher\"..." runtime=io.containerd.runc.v2 type=io.containerd.ev
Feb 07 14:35:29 box1-t112 dockerd[3216]: ttime="2025-02-07T14:35:29.030164400Z" level=info msg="loading plugin \"io.containerd.internal.v1.shutdown\"..." runtime=io.containerd.runc.v2 type=io.containerd.i
Feb 07 14:35:29 box1-t112 dockerd[3216]: ttime="2025-02-07T14:35:29.030116193Z" level=info msg="loading plugin \"io.containerd.trrpc.v1.task\"..." runtime=io.containerd.runc.v2 type=io.containerd.trrpc.v1
Feb 07 14:35:29 box1-t112 dockerd[3216]: ttime="2025-02-07T14:35:29.030201167Z" level=info msg="starting signal loop" namespace=moby path=/var/tmp/docker_exec/containerd/daemon/io.containerd.runtime.v2.1

root@box1-t112:/home/developer# docker ps -a
CONTAINER ID        IMAGE                                     COMMAND                  CREATED             STATUS              PORTS                               NAMES
19382648fc13        localhost:5000/docker-registry-01.nomadroll.com/nd-portal/nde:202304-10.1.7-x64  "/nde/run.sh"           38 minutes ago      Up 38 minutes (healthy)            0.0.0.0:443->443/tcp            portal
1604ddb87dc5        registry:2                             "/entrypoint.sh /etc..." 39 minutes ago      Up 39 minutes              127.0.0.1:5000->5000/tcp            registry
```

6.3.9.2. Portal image starts and reports healthy

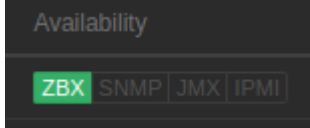
Test ID:	6.3.9.4
Description:	Ensure that after the docker container for the portal automatically starts
Execution Steps:	8. Reboot the CCU 9. Access the CCU via console or SSH 10. Once on the CCU terminal escalate privileges using ‘sudo su’ 11. Confirm that the docker is running using ‘docker ps’ 12. Wait for the container portal to start should have the status ‘up <time> (starting)’ 13. If the container reports unhealthy, wait thirty minutes as the portal could be syncing Also ensure the CCU is <i>up-to-date</i> <i>Perform powercycle if required</i> 1. Check docker ps again for the image reporting healthy
Expected Outcome:	After the CCU is rebooted/power cycled the CCU will start the docker container ‘portal’, the portal container will be in state healthy. If the container has existed for a short time, it may require time for the container to enter this state and may even require a reboot
Preconditions:	Base NC image is sync’d with puppet Access through SSH or serial to CCU Previous docker tests passed (6.3.9.1) MAR tunnel is up Docker service automatically starts

Bench:	Devops bench
Test Method:	Functional Testing
Test Device	Linux
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz
Evidence:	
<pre>root@box1-t112:/home/developer# docker ps -a CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES 19382648fc13 localhost:5000/docker-registry-01.nonadrail.com/nd-portal/nde:2023Q4-10.1.7-x64 "/nde/run.sh" 39 minutes ago Up 39 minutes (healthy) 0.0.0.0:443->443/tcp portal 1604ddb87dc registry:2 "/entrypoint.sh /etc..." 41 minutes ago Up 41 minutes 127.0.0.1:5000->5000/tcp registry root@box1-t112:/home/developer#</pre>	

6.3.10. Reporting

6.3.10.1. MQTT functionality

Test ID:	6.3.10.1
Description:	MQTT reporting is working as expected in the form of the telemetry process and is reporting to the backend MQTT server
Execution Steps:	Ensure unit is online and tunnel is connected to backend confirm with '<i>marcli</i>' Open mqtt explorer (or other mqtt software) and connect to backend MQTT server Wait some time to allow data to be sent to HiveMQ server Ensure that data is reporting for unit under the following structure: (PROJECT> TRAIN> CCU> DATA FIELDS) p39>113>u1>DATA Ensure that data for the nomad dictionary is also reporting (bq reporting) CUSTOMER>FLEET>FLEET>UNIT>NULL>CCU>DATA ALS018/norsketog72/norsketog72/113/null/CCU1/DATA
Expected Outcome:	Telemetry and other relevant data are being forwarded to the MQTT broker and are appearing under the correct structure. This data meets the Nomad dictionary standards.
Preconditions:	Unit is sync'd with puppet, unit is connected to MAR3 backend Test has been granted access to backend MQTT server Tester has a method of reading MQTT messages (such as mqtt explorer)
Bench:	Devops bench
Test Method:	Functional Testing
Test Device	Linux
Execution Date	07/02/2025
Executer	Szymon Dutkiewicz
Evidence:	

	Go to the administration tab and select proxies Ensure that the proxy is reporting to the Zabbix server Go to configuration> hosts In the name field search the train name Ensure that for all results returned the Availability column shows at least one square green (see image) 
Expected Outcome:	The units Zabbix proxy is reporting to the server The devices linked to the proxy are available to Zabbix either via snmp or Zabbix agent (excluding portal)
Preconditions:	CCU is sync'd with puppet backend CCU is online and is connected to the backend Tester has access to Zabbix Unit has been manually added to Zabbix OR added through NMS
Bench:	Devops bench
Test Method:	Functional Testing
Test Device:	N/A
Execution Date:	19/02/2025
Executer:	Alex Gabbatiss
Evidence: Proxy is reporting into zabbix	

Preconditions:	Unit is connected to MAR backend Test executer has NMS access Unit is in NMS or can be added Unit is reporting valid GPS on CCU
Bench:	Devops bench
Test Method:	Functional testing
Test Device:	N/A
Execution Date:	19/02/2025
Executer:	Szymon Dudkiewicz

Evidence:

Confirmed that Unit was reporting valid GPS data

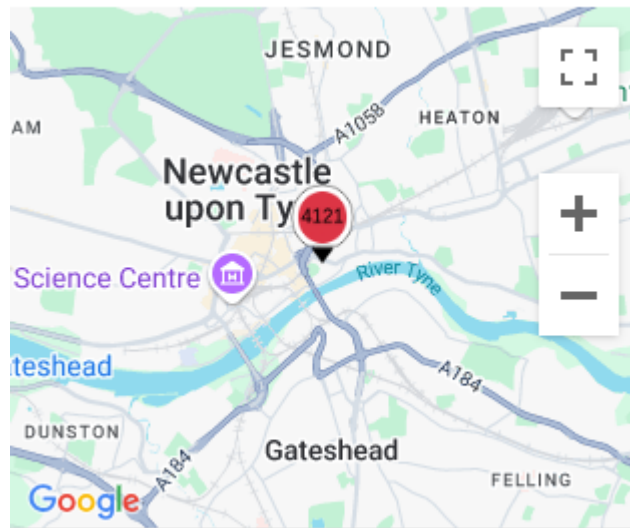
```

root@box1-t121:/home/developer# marcli geo
GPS updates enabled:      TRUE
Last update:      Wed Feb 19 14:11:39 UTC 2025

Valid:      TRUE
Latitude:      54.97043
Longitude:      -1.60492
Altitude:      39.12900
Heading:      0.00
Speed:      0 km/h
root@box1-t121:/home/developer#

```

In NMS, the unit is reporting its accurate location



6.3.10.4. Fault Reporting in NMS

Test ID:	6.3.10.4
Description:	Ensure that when a fault on the CCU (such as a failed modem or missing AP) is reported in Zabbix, it is also reported in NMS for that unit
Execution Steps	Log into Zabbix/NMS confirm active faults for test bench Main page> SECTION 'Problems by severity' > Bench number Create fault on unit (such as changing device config or powering device down) Wait for fault to be picked up by zabbix/NMS Check zabbix for fault Check NMS for fault Resolve fault Check that fault disappears in NMS and Zabbix
Expected Outcome:	Test is passed when faults presented in Zabbix are also present in NMS and subsequently cleared.

Preconditions:	Unit is connected to backend via tunnel Test executer has NMS access Unit is in NMS or can be added Unit is reporting valid GPS on CCU
Bench:	Devops bench
Test Method:	Functional testing
Test Device:	N/A
Execution Date:	19/02/2025
Executer:	Alex Gabbatiss

Evidence:

Curent faults in NMS

nms-oebb.nomadrail.com/nms/#/project/50/monitoring-detail/train/4121/summary						
Clear selected alarms Investigate in Zabbix Acknowledge selected alarms Acknowledge all alarms						
☐	Train	Severity	Device	Description	Date	Acknowledged
	⌵	⌵	⌵	⌵	⌴	⌵
		All ▾				All ▾
☒	4121 (4121)	High	C2 AP4	Access point 7.7.7.7 cannot be pinged	14/02/25 12:10	✖
☒	4121 (4121)	High	C2 AP3	Access point 7.7.7.7 cannot be pinged	14/02/25 12:10	✖
☒	4121 (4121)	High	C2 AP2	Access point 7.7.7.7 cannot be pinged	14/02/25 12:10	✖
☒	4121 (4121)	High	C2 AP1	Access point 7.7.7.7 cannot be pinged	14/02/25 12:10	✖
☒	4121 (4121)	High	C2 CCU1	Potential Split brain issue	14/02/25 12:20	✖
☐	4121 (4121)	Low	C2 CCU1	modem 1 not active	14/02/25 12:28	✖
☐	4121 (4121)	Low	C2 CCU1	modem 2 not active	14/02/25 12:28	✖
☐	4121 (4121)	Low	C2 CCU1	modem 3 not active	14/02/25 12:29	✖
☐	4121 (4121)	Low	C2 CCU1	Modem 0 has a missing SIM. Follow procedure (Confluence) and report this.	14/02/25 12:36	✖
☐	4121 (4121)	Low	C2 CCU1	Modem 1 has a missing SIM. Follow procedure (Confluence) and report this.	14/02/25 12:36	✖
☐	4121 (4121)	Low	C2 CCU1	Modem 2 has a missing SIM. Follow procedure (Confluence) and report this.	14/02/25 12:36	✖
☐	4121 (4121)	Low	C2 CCU1	Modem 3 has a missing SIM. Follow procedure (Confluence) and report this.	14/02/25 12:37	✖
☒	4121 (4121)	High	C1 AP4	Access point 7.7.7.7 cannot be pinged	19/02/25 13:23	✖
☒	4121 (4121)	High	C1 AP3	Access point 7.7.7.7 cannot be pinged	19/02/25 13:23	✖
☒	4121 (4121)	High	C1 AP2	Access point 7.7.7.7 cannot be pinged	19/02/25 13:23	✖
☒	4121 (4121)	High	C1 AP1	Access point 7.7.7.7 cannot be pinged	19/02/25 13:23	✖
☒	4121 (4121)	High	C3 AP4	Access point 7.7.7.7 cannot be pinged	19/02/25 13:23	✖
☒	4121 (4121)	High	C3 AP3	Access point 7.7.7.7 cannot be pinged	19/02/25 13:23	✖
☒	4121 (4121)	High	C3 AP2	Access point 7.7.7.7 cannot be pinged	19/02/25 13:23	✖
☒	4121 (4121)	High	C3 AP1	Access point 7.7.7.7 cannot be pinged	19/02/25 13:23	✖
☒	4121 (4121)	High	C4 AP4	Access point 7.7.7.7 cannot be pinged	19/02/25 13:24	✖
☒	4121 (4121)	High	C4 AP3	Access point 7.7.7.7 cannot be pinged	19/02/25 13:24	✖
☒	4121 (4121)	High	C4 AP2	Access point 7.7.7.7 cannot be pinged	19/02/25 13:24	✖

CCU Fault test

Removed GPS cable from CCU to generate GPS reporting fault

Waited 90 minutes as this is the time needed to generate a GPS fault

4121 (4121)	Low	C2 CCU1	Invalid GPS for 90 minutes (under cover?)	19/02/25 16:51	✕
-------------	-----	---------	---	----------------	---

Re-connected GPS cable

After some time, the fault disappeared from NMS and the fault appeared resolved in zabbix

2025-02-19 17:51:55 ☐ Warning 10:27:04 RESOLVED 50 4121_MAR3-B1 Invalid GPS for 90 minutes (under cover?)

AP fault test

The NCL bench currently only has one active AP

nms-oebb.nomadrail.com/nms/#/project/50/monitoring-detail/train/4121/summary

Location	Next trip	Speed	Users	Links	Down	Up	Checked	Last message	Last online
AC 516km	1	8.52 km/h	B1 (0)	B1M0 B1M1 B1M2 B1M3 B1ETH B1WIFI	0.04	0.06	15:51	19/02/2025 15:50	19/02/2025 15:50

A

G

E

B

☐	Train	Severity	Device	Description	Date	Acknowledged
☐	4121 (4121)	High	C2 AP4	Access point 7.7.7.7 cannot be pinged	14/02/25 12:10	✕

AP unplugged from switch

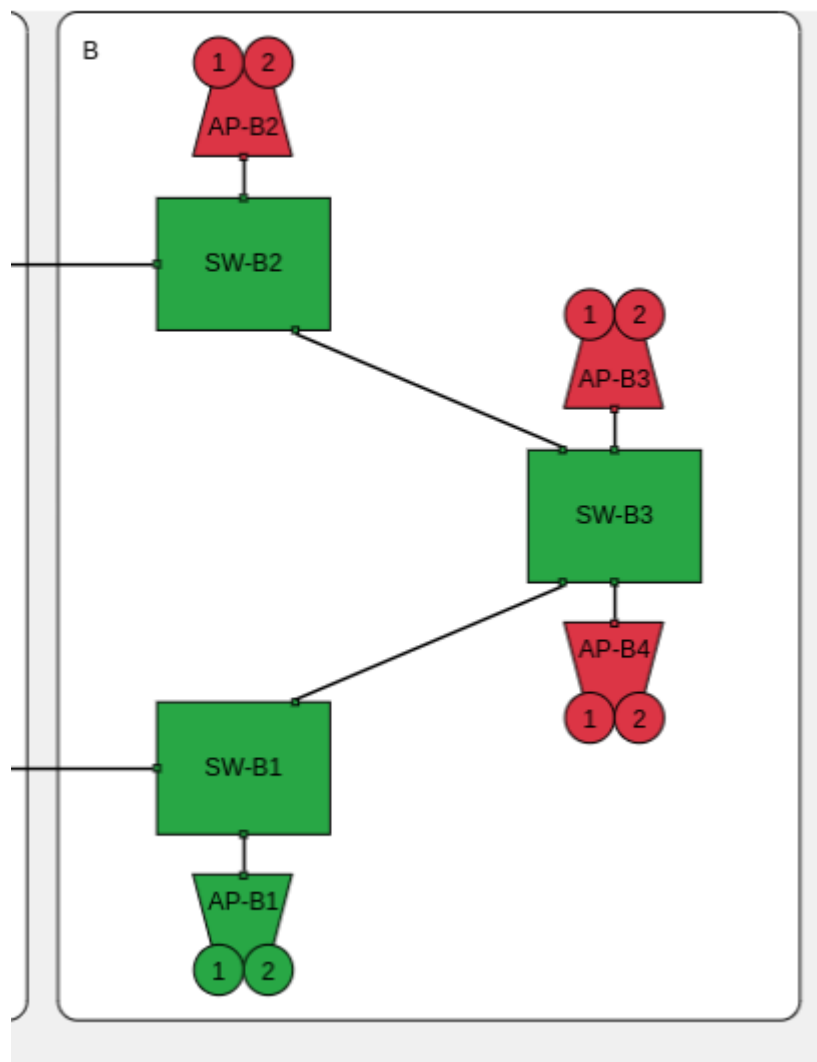
Waited a few minutes

AP began reporting down

4121 (4121)	High	C4 AP1	Access point 7.7.7.7 cannot be pinged	19/02/25 15:57	✕
-------------	------	--------	---------------------------------------	----------------	---

AP powered back up

After five minutes the AP began reporting into NMS as green



7. Test Completion Overview & Recommendations

The functional requirements were gathered based on project specific requirements. All functional testing was completed successfully on local bench in Newcastle, the in-country testing to be completed at a later date. Testing carried out returned successful result for each of the test ran. However, it should be noted again, that the Newcastle bench replicated the 4-coach train setup, the 6-coach set up was not tested, although checks have been taken to ensure that the 6-coach setup works as expected.